32

Antineutrophil Cytoplasmic Antibody–Associated Vasculitis and Antiglomerular Basement Membrane Disease

Andreas Kronbichler | Martin Windpessl | David R.W. Jayne | Ingeborg M. Bajema

CHAPTER OUTLINE

KEY POINTS

- Antineutrophil cytoplasmic antibody glomerulonephritis is present in 70% of AAV patients, while kidney involvement in antiglomerular basement membrane disease is almost 100%.
- Antiglomerular basement membrane disease is the most aggressive autoimmune disorder with a poor kidney prognosis.
- Novel therapeutic approaches in antineutrophil cytoplasmic antibody glomerulonephritis have transformed the disease from a fatal to a chronic relapsing-remitting disease.
- Kidney biopsies are required in most patients beyond establishing a diagnosis but are used for prognostication.

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS

Both antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis and anti–glomerular basement membrane (GBM) disease have traditionally been categorized as diseases belonging to the spectrum of "rapidly progressive glomerulonephritis" (RPGN). RPGN has been defined histologically by the presence of at least 50% (cellular) crescents on a kidney biopsy and clinically by a rapid deterioration of kidney function and features of nephritis, characterized by hematuria of glomerular origin and proteinuria.[1] Classification of RPGN is based on the pattern of immunoglobulin deposition detected by immunofluorescence in patients with crescentic glomerulonephritis (Table 32.1).[2]

The "RPGN" classification is somewhat outdated due to several reasons: 1. Most patients with ANCA-glomerulonephritis (GN), the most common RPGN form, will not present with a crescentic class according to the Berden classification. In a real-life study, only 25.5% of patients presented with a crescentic glomerulonephritis[3]; 2. The kinetics of kidney function decline are often unavailable due to the lack of prior kidney function tests, and a proportion of patients, especially those with myeloperoxidase (MPO)-ANCA vasculitis, presents with a slowly progressive form of ANCA-GN[4]; 3. Patients increasingly present with normal kidney function or only mild elevation of serum creatinine,[5] which has also been reported for pediatric cases of anti-GBM disease,[6] likely due to an increase in awareness to consider these diseases as differential diagnosis; and 4. The presence of crescents in biopsies taken from IgA nephropathy patients had not improved the accuracy of the IgA nephropathy (IgAN) Prediction Tool at 12 and 24 months post kidney biopsy.[7] Thus neither the severity of kidney failure at baseline nor the presence of cellular crescents are accurate measures to predict acuity of an autoimmunity-related GN.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.1 Overview of Types, Associated Patterns, Immunofluorescence Findings, and Associated Diseases in Presence of "Rapidly Progressive Glomerulonephritis"

Type	Pattern	Immunofluorescence	Disease(s)
Type 1	Linear	Linear deposits of immunoglobulin G with in situ immune-complex formation	Anti-GBM disease
Type 2	Granular	Granular deposits of immunoglobulin	Lupus nephritis (classes III and IV), IgA vasculitis/ IgA nephropathy, postinfectious glomerulonephritis, cryoglobulinemic vasculitis, MPGN (type I and II), fibrillary glomerulonephritis, membranous nephropathy
Type 3	Pauci-immune	"Virtual" absence of glomerular immune deposits	ANCA-glomerulonephritis

ANCA, Antineutrophil cytoplasmic antibody; *GBM,* glomerular basement membrane; *MPGN,* membranoproliferative glomerulonephritis.

This chapter therefore focuses on the disease entities ANCA-associated vasculitis and anti-GBM disease rather than providing an overview over other causes of RPGN, as both are forms of acute glomerulonephritis that will progress to kidney failure without appropriate therapy.

ANTINEUTROPHIL CYTOPLASMIC ANTIBODY–ASSOCIATED VASCULITIS

INTRODUCTION

Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) encompasses three distinct disease phenotypes: granulomatosis with polyangiitis (GPA), microscopic polyangiitis (MPA), and eosinophilic granulomatosis with polyangiitis (EGPA).[8] Historically, GPA and MPA have been studied together in clinical trials and observational studies, while cases with EGPA have been studied separately. The reasons for this become apparent when looking at disease manifestations among GPA, MPA, and EGPA: Approximately 80% of patients with MPA and 60% with GPA present with kidney disease. The frequency of kidney involvement in EGPA is considerably lower; it has been estimated around 25%,[9] but this estimate may be influenced by recruitment bias and the true number seems to be even lower.

A diagnosis of AAV is substantiated by the presence of ANCA. Antibody testing has moved away from testing for the presence of cytoplasmic or perinuclear ANCA by immunofluorescence to specific immunoassays that allow detection of ANCA directed either against proteinase 3 (PR3) or myeloperoxidase (MPO), and these tests have higher specificity and sensitivity to support a diagnosis of AAV.[10] PR3-ANCA is usually detected in cases with GPA, while MPO-ANCA is more frequent in MPA. Severe systemic presentation forms, such as patients with kidney disease, usually have detectable antibodies. This is different in patients with EGPA who present with ANCA-positivity in only 30% of cases. Antibody-positive patients (more often MPO-ANCA than PR3-ANCA–positive) have a higher frequency of kidney disease (29%) in comparison with those who are ANCA-negative (9%). The EGPA patient subgroup with MPO-ANCA–positivity shares a genetic association at the major histocompatibility complex (MHC) class II *HLA-DQ* locus with MPO-ANCA–positive MPA,[11] thus in part explaining some of the observed similarities in presentation of MPA and MPO-positive EGPA. For further discussions in this chapter, we focus on GPA and MPA; however, kidney disease in EGPA requires the same/similar management as discussed later. Specific manifestations in proposed disease clusters, also highlighting extrarenal disease, see Table 32.2.[12] Major differences are encountered depending on diverse presentation forms, and kidney involvement in general is associated with poor overall prognosis.

Once a lethal disease with a 5-year mortality rate of 87% in the absence of treatment,[13] in the more recent era (2010–2015) the 5-year mortality of AAV has decreased to below 20%[14] and AAV has become a relapsing-remitting disease due to refined therapeutic approaches. Both GPA and MPA are systemic diseases that require intensive immunosuppressive therapy to prevent organ damage from vasculitic manifestations, such as alveolar hemorrhage, ANCA-glomerulonephritis, and neuropathy. These immunosuppressive therapies have a substantial toxicity potential, which leads to a significant risk of infections within the first year after diagnosis, and together with the sequelae of the disease, such as CKD, immunosuppressants are also drivers of an increased event rate of cardiovascular complications.[15,16] The burden of the disease itself, together with the detrimental effects of immunosuppression, still leads to an overall 2.3-fold higher mortality risk of patients with AAV in comparison with a matched background population.[17] Efforts to rapidly and effectively control the disease but also limit exposure to and complications associated with immunosuppressive therapies such as glucocorticoids are therefore both necessary to reduce premature death among patients with AAV. Novel therapeutic approaches and regimens have been approved in the management of GPA and MPA and might help to achieve this goal.[18]

SEROTYPE VERSUS PHENOTYPE FOR DIAGNOSIS

There is ongoing debate whether patients should be classified on the basis of their ANCA serotype rather than their clinical phenotype. The ANCA subtypes of MPO-ANCA and PR3-ANCA correspond in high frequencies to the diagnoses of MPA and GPA, although a proportion of patients, with either a more limited disease form without kidney disease

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.2 Cluster Analysis of 3868 Patients with Antineutrophil Cytoplasmic Antibody–Associated Vasculitis, Derived from 6 European Registries

	Severe Kidney	MPO-Kidney	PR3-Kidney	Young Respiratory Cluster	Inflammatory Multisystem Cluster
Patients	555 (14.3%)	782 (20.2%)	683 (17.7%)	646 (16.7%)	1202 (31.1%)
Age	67	63	56	48	55
Follow-up (years)	3	3	4	6	5
Women	44%	50%	36%	59%	50%
MPA	62%	76%	17%	18%	22%
Organs affected	3	2	5	3	4
Constitutional	61%	34%	89%	39%	72%
Musculoskeletal	44%	31%	80%	43%	65%
Skin	7%	12%	42%	21%	30%
ENT	20%	14%	70%	70%	60%
Lung	52%	35%	67%	42%	57%
Cardiovascular	8%	4%	13%	6%	7%
Gastrointestinal	6%	3%	17%	3%	7%
CNS	2%	2%	10%	9%	12%
PNS	9%	4%	25%	15%	28%
Kidney	100%	99%	99%	23%	37%
ANCA	37%	17%	82%	60%	68%
PR3-ANCA	58%	75%	13%	22%	24%
MPO-ANCA					
Creatinine (μmol/L; mg/dL)	580; 6.6	240; 2.7	221; 2.5	80; 0.9	79; 0.9
CRP (mg/L)	123	16	110	6	99
Death	31%	21%	23%	6%	15%
KF	42%	28%	20%	2%	4%

Data-driven subclassification revealed the presence of 5 clusters,[12] with highly variable clinical manifestations. Most patients with kidney disease present in clusters associated with acute kidney disease at baseline (severe kidney, myeloperoxidase, and proteinase 3-ANCA clusters). These clusters are associated with higher risks of kidney failure and death. The spectrum of extrarenal disease manifestations and inflammation at baseline is also variable.

ANCA, Antineutrophil cytoplasmic antibody; *CNS,* central nervous system; *CRP*, C-reactive protein; *ENT*, ear, nose, and throat; *KF,* kidney failure; *MPA,* microscopic polyangiitis; *MPO,* myeloperoxidase; *PNS,* peripheral nervous system; *PR3,* proteinase 3.

or a subset with "pauci-immune" glomerulonephritis, which is often renal limited, remains ANCA-negative. A diagnosis in ANCA-negative disease is dependent on clinical presentation and biopsy findings. The revised classification criteria for GPA and MPA assigned specific weight to the presence of ANCA (see Table 32.3), further complicating the path to a potential diagnosis of AAV in those with no detectable ANCA when a vasculitis is suspected.

A meta-analysis of genetic associations of patients with ANCA-associated vasculitis revealed that 75% of the associations are more closely related to PR3 and MPO rather than the diagnosis of GPA and MPA.[19] A clinical study focusing on pulmonary presentation forms identified that some features are exclusively observed in PR3-ANCA vasculitis, such as pulmonary nodules, and MPO-ANCA vasculitis (Fig. 32.1), such as an usual interstitial pneumonia pattern,[20] the regulation of different biomarkers,[21] the prediction of disease relapses as patients with PR3-ANCA have a higher relapse risk than those with MPO-ANCA,[22] and also the occurrence of certain comorbidities, with a predominance of cardiovascular mortality in MPO-ANCA–positive patients.[17] While most clinical trials have required a clinical diagnosis of GPA or MPA to qualify for trial inclusion, the Plasma Exchange and Glucocorticoids in Severe ANCA-Associated Vasculitis (PEXIVAS) trial randomized patients in a 2 × 2 factorial design (plasma exchange (PLEX) versus no-PLEX and reduced-dose versus standard-dose glucocorticoids; see discussions later) according to their ANCA serotype (historical or current positivity for MPO- or PR3-ANCA)[23] and smaller experimental medicine studies, such as the ongoing ObiVas trial (ISRCTN13069630), a trial investigating the endpoint of nasal-associated lymphoid tissue $CD19^+$ B cell depletion in obinutuzumab- versus rituximab-treated patients, required PR3-ANCA–positivity for inclusion.[24] The importance of a positive ANCA-test has also been highlighted by the recent American College of Rheumatology (ACR)/European Alliance of Associations for Rheumatology (EULAR) classification criteria for GPA and MPA: A positive PR3-ANCA (and/or cytoplasmic ANCA) and MPO-ANCA (and/or perinuclear ANCA) would be sufficient to classify a patient as having GPA and MPA in the context of suspected vasculitis.[25,26] On the basis of the previously mentioned considerations, it is important to report the ANCA serotype and the clinical phenotype when reporting cases of AAV (Table 32.4), as has also been agreed on in the 2012 revised international Chapel Hill Consensus Conference nomenclature of vasculitides.[8]

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.3 Classification Criteria for Granulomatosis with Polyangiitis and Microscopic Polyangiitis as Issued by the American College of Rheumatology and European Alliance of Associations for Rheumatology

	2022 ACR/EULAR Classification Criteria for GPA	2022 ACR/EULAR Classification Criteria for MPA
Criteria	A total cutoff of ≥5 is needed. Clinical criteria: • Nasal involvement: bloody discharge, ulcers, crusting, congestion, blockage, or septal defect/perforation (+3) • Cartilaginous involvement (inflammation of ear or nose cartilage, hoarse voice or stridor, endobronchial involvement, or saddle nose deformity) (+2) • Conductive or sensorineural hearing loss (+1) Laboratory, imaging, and biopsy criteria: • Positive test for cANCA or anti-PR3 antibodies (+5) • Pulmonary nodules, mass, or cavitation on chest imaging (+2) • Granuloma, extravascular granulomatous inflammation, or giant cells on biopsy (+2) • Inflammation, consolidation, or effusion of the nasal/paranasal sinuses, or mastoiditis on imaging (+1) • Pauci-immune glomerulonephritis on biopsy (+1) • Positive test for pANCA or anti-MPO antibodies (–1) • Blood eosinophil count ≥1 x 10^9/L (–4)	A total cutoff of ≥5 is needed. Clinical criteria: • Nasal involvement: bloody discharge, ulcers, crusting, congestion, blockage, or septal defect/perforation (-3) Laboratory, imaging, and biopsy criteria: • Positive test for pANCA or anti-MPO antibodies (+6) • Fibrosis or interstitial lung disease on chest imaging (+3) • Pauci-immune glomerulonephritis on biopsy (+3) • Positive test for cANCA or anti-PR3 antibodies (-1) • Blood eosinophil count ≥ 1x10^9/liter (-4)
Patients/Controls (Comparators)	Development set: 578/652 Validation set: 146/161	Development set: 149/408 Validation set: 142/414
Validation	Sensitivity of 92.5% and specificity of 93.8% in the validation cohort (ACR 1990 in same dataset: sensitivity 69.3%, specificity 75.8%)	Sensitivity of 90.8% and specificity of 94.2% in the validation cohort (No ACR classification criteria existed for MPA)

ACR, American College of Rheumatology; *ANCA,* antineutrophil cytoplasmic antibody; *EULAR*, European Alliance of Associations for Rheumatology; *MPA*, microscopic polyangiitis; *MPO,* myeloperoxidase; *PR3,* proteinase 3.

EPIDEMIOLOGY

The global incidence of GPA and MPA has increased over the past decades. This is in part explainable by a pronounced increase in ANCA testing, which has become available in most laboratories worldwide. In a comparison between two time periods in Denmark (2000–2004 versus 2010–2015a, while testing increased 3-fold over time, the incidence of GPA and MPA increased from 15.1 to 21.4 per million.[14] Even higher incidence rates were reported in other studies focusing on epidemiology of the disease. In Olmsted County, Minnesota, in a time period from 1996 to 2015, the incidence of GPA was 13 cases per million and of MPA was 16 cases per million, with a slight male preponderance. Most studies reporting patients with GPA found a male predominance, while many focusing on MPA found a female preponderance.[30] Patients with GPA were on average a decade younger as patients with MPA at the time of diagnosis.[31] Further studies on epidemiology of AAV have been reviewed elsewhere.[32]

Geoepidemiologic differences have been reported in AAV. In a study of 967 patients, those from Japan and China have an almost 60- and 7-times greater chance of being MPO-ANCA rather than PR3-ANCA–positive than patients from Northern Europe. In addition, differences in Europe or of individuals with European ancestry have also been observed. Patients in Southern Europe and Caucasian Americans had a 2.6-fold greater chance of MPO-ANCA–positivity than Northern Europeans.[33] This phenomenon has been linked to northern latitudes and lower ultraviolet radiation levels, as both were associated with PR3-ANCA–positivity in a study of 1408 patients from Europe and the United States.[34] These findings further highlighted a north-south dichotomy for patients with a positive PR3-ANCA test, but the predominance of MPO-ANCA in Eastern Asia remains largely unexplained thus far. In the European cluster analysis, patients in the MPO-kidney cluster were on average 63 years old, while those within the PR3-kidney cluster were 56 years old. The young respiratory cluster, also dominated by cases with PR3-ANCA vasculitis, were predominantly women, while the other clusters had either more men or an equivalent distribution (see Table 32.2).[12]

PATHOGENESIS OF RENAL INJURY

In diverse autoimmune diseases a loss of T cell and B cell tolerance leads to autoimmunity, which precedes an inflammatory response and tissue injury. In the case of AAV, there is a loss of immunologic tolerance to the neutrophil enzymes MPO and/or PR3. Subsequently, ANCA will attach to these proteins promoting neutrophil activation. Occurrence of tissue injury is mediated by both humoral and cellular autoimmunity and involves multiple innate and adaptive components of the immune system (Fig. 32.2).[35] ANCAs are often detectable at a time before the diagnosis of

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

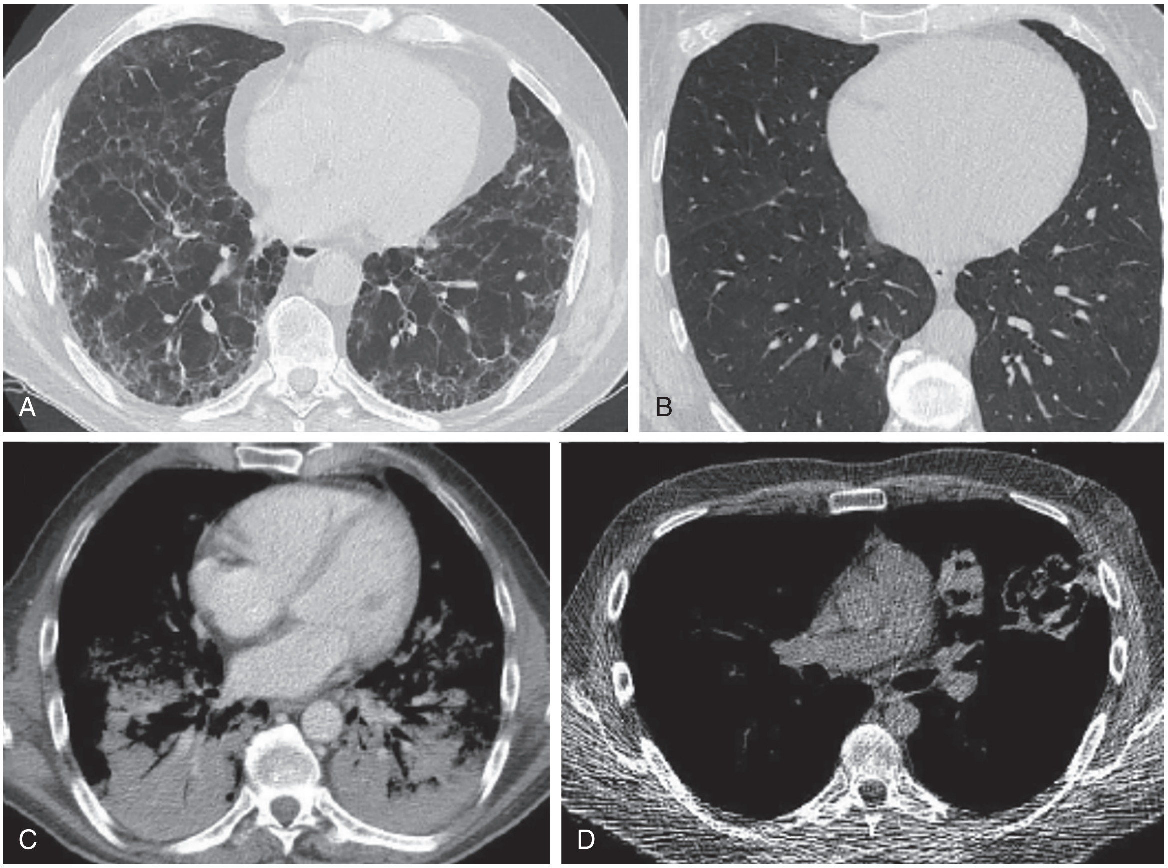

Fig. 32.1 Different patterns of lung involvement encountered in patients with antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis. Computed tomography scan images of various forms of lung injury in patients with ANCA-associated vasculitis. (A) Honeycombing and peripheral reticulation, usual interstitial pneumonitis, as exclusively observed in patients with myeloperoxidase-ANCA vasculitis. (B) Mild ground-glass opacities (diffuse) in a perivascular pattern, here in a patient with myeloperoxidase-ANCA vasculitis. (C) Diffuse alveolar hemorrhage in a patient with proteinase 3-ANCA vasculitis. (D) Two large granulomas in a patient with proteinase 3-ANCA vasculitis.

AAV, often years ahead of diagnosis, was made.[36,37] Disease activity includes granulomatous manifestations, which are almost exclusively seen in patients with PR3-ANCA vasculitis but rarely manifest in the kidneys, and vasculitis manifestations, which frequently lead to life-threatening disease such as alveolar hemorrhage and kidney failure and seen in patients with either PR3-ANCA or MPO-ANCA.

Multinucleated giant cell formation and development of granulomas in GPA are promoted by stimulation of monocytes by PR3, and this depends on soluble interleukin-6, monocyte MAC-1, and protease-activated receptor-2.[38] Vasculitis is characterized by microvascular endothelial inflammation, which in turn leads to extravascular inflammation, tissue destruction, fibrotic/sclerotic lesions, and eventually loss of tissue function. Key effector cells leading to tissue destruction are neutrophils, monocytes, and macrophages, all capable of either expressing or producing the key autoantigens, namely PR3 and MPO. Key inflammatory pathways induced by both antigens lead to endothelial injury through mechanisms such as neutrophil degranulation and microparticle release from neutrophils and are major constituents of neutrophil extracellular traps (NETs), networks of fibers composed of DNA from neutrophils relevant in antimicrobial processes,[39] and contributors to endothelial cell damage directly through induction of proinflammatory proteins and activation of the complement system.[40]

The alternative complement pathway plays a crucial role in the pathogenesis of ANCA-GN. Activation of the complement system contributes to further neutrophil-mediated inflammation via production of anaphylatoxins, such as complement C5a. Seminal work in 2007 demonstrated that the injection of anti-MPO IgG was capable of inducing ANCA-GN in $C4^{-/-}$ mice, while $C5^{-/-}$ and factor $B^{-/-}$ mice were protected and did not develop cellular crescents or necrosis.[41] These findings indicated that the alternative complement pathway is required to induce renal injury in ANCA-GN, while the classic and lectin pathways are not. Likewise, pretreatment of mice with a C5-inhibiting monoclonal antibody prevented development of crescents and fibrinoid necrosis and led to a significant reduction of influx of inflammatory cells.[42] Further experiments provided evidence that the addition of CCX168 (now known as avacopan, a drug being used clinically) at day 1 significantly reduced the development of cellular crescents by 89.2%, which followed a dose-dependent pattern, and reduced hematuria and proteinuria in treated animals.[43] Blockade of C5L2, the second receptor of C5a, had no relevant effect on blocking renal inflammation in this MPO-ANCA mouse model. One study induced lung injury in an

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.4 Disease Manifestations in GPA, MPA, PR3-ANCA, and MPO-ANCA Vasculitis

	GPA	MPA	PR3-ANCA Vasculitis	MPO-ANCA Vasculitis
General	78%	86%	81%	92%
Fever ≥38°C (100.4°F)	31%	40%	44% (≥38.5°C [101.3°F])	
Fatigue	56%	68%	—	
Weight loss ≥2 kg (4.4 lb)	35%	43%	47% (>3 kg [6.6 lb])	
Arthralgia	55%	32%	56%	
Myalgia	22%	24%	26%	
Cutaneous	35%	30%	34%	17%
Petechiae or purpura	17%	10%	18%	
Mucous membranes/Eyes	38%	13%	28%	10%
Scleritis or episcleritis	14%	1%	5% and 10%	
ENT	82%	26%	81%	2%
Respiratory	63%	63%	68%	50%
Hemoptysis/DAH	21%	19%	18%	22%
Cardiovascular	11%	15%	16%	6%
Abdominal	19%	22%	11%	4%
Renal	59%	82%	58%	79%
Neurologic	31%	37%	30%	39%
Neuropathy	12%	26%	21%	21%
Mononeuritis multiplex	5%	9%	—	
Sensory neuropathy	11%	21%	—	

The frequencies are taken as reported in 674 patients with granulomatosis with polyangiitis (GPA) and 325 patients with microscopic polyangiitis reported in the Diagnostic and Classification Criteria in Vasculitis study,[9] an international study involving 32 countries, and 546 patients with PR3-ANCA vasculitis reported from the French Vasculitis Study Group.[28] The cohort of patients with MPO-ANCA vasculitis (*n* = 144) also comprises 6 cases with PR3-ANCA vasculitis and reported data from a single center in Germany.[29]

ANCA, Antineutrophil cytoplasmic antibody; *DAH,* diffuse alveolar hemorrhage; *DCVAS,* Diagnostic and Classification Criteria in Vasculitis; *ENT,* ear, nose and throat; *FVSG,* French Vasculitis Study Group; *GPA,* granulomatosis with polyangiitis; *MPA,* microscopic polyangiitis; *MPO,* myeloperoxidase; *PR3,* proteinase 3.

animal model of MPO-ANCA vasculitis, either mimicking GPA, MPA, or EGPA. All of the animals developed crescentic GN, which was successfully inhibited by knocking out C5 or factor B, as was shown previously. However, blockade of the alternative complement pathway did not reduce the lung injury score,[44] underlining that induction of lung injury in AAV might not depend on activation of the alternative complement pathway.

Immunofluorescence (IF) or immunohistochemistry (IHC) findings of kidney specimen in ANCA-GN are typically “pauci-immune,” which means that immune deposits are either absent or only faintly present on biopsy, although deposition of at least one immunoglobulin or complement component has been observed in over half of the renal biopsies in one series[45] (discussed further “Histology” later).

Macrophages and monocytes play key roles in the kidney injury observed in ANCA-GN. CD163 is a high-affinity scavenger receptor for the hemoglobin-haptoglobin complex that is exclusively expressed on monocytes and macrophages. Urinary levels of soluble CD163 (sCD163) are elevated during phases of active vasculitis, and the expression levels are higher in comparison with patients with sepsis-associated acute kidney injury or patients with other forms of glomerulonephritis, indicating a dominant role of monocytes and macrophages in propagating kidney injury in ANCA-GN in comparison with other diseases leading to kidney failure. Glomerular and tubulointerstitial expression of CD163 is more pronounced in comparison with other glomerular diseases,[46] further highlighting the relevance of CD163 in ANCA-GN. Importantly, the levels of urinary sCD163 tend to correlate with severity of kidney injury, in particular fibrinoid necrosis and cellular crescents.[47] Sequential measurement of urinary sCD163 predicted disease relapse better than measuring proteinuria, hematuria, or repeat kidney biopsy,[47,48] although the specificity of urinary sCD163 was reduced when nephrotic-range proteinuria was present during remission.[48]

Various other mediators lead to a mixed inflammatory infiltrate observed in ANCA-GN. Monocyte chemoattractant protein-1 (MCP-1; also known as chemokine ligand 2, CCL2), is a chemokine that recruits monocytes, memory T cells, and dendritic cells to sites of inflammation. CCR2 and CCR4 are two cell surface CC chemokine receptors that bind CCL2. MCP-1 was elevated in urine samples of patients with active ANCA-GN, and a reduction of urinary MCP-1 levels preceded improvement in kidney function.[49] Longitudinal follow-up sampling of patients revealed that the addition of urinary MCP-1 to urinary sCD163 improved the accuracy of identifying subtle ANCA-GN flares.[50] Urinary MCP-1/creatinine ratio has been tested in the phase 2 CLEAR trial, testing two different avacopan regimens (without glucocorticoids and with 20 mg prednisone) against a standard of care cohort receiving 60 mg prednisone; notably, patients received either cyclophosphamide (79.4%) or rituximab (20.6%) alongside avacopan or prednisone. Both avacopan-treated cohorts had a more rapid reduction in urinary MCP-1 levels in comparison with the prednisone arm.[51] CC chemokine ligand 18 (CCL18),

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

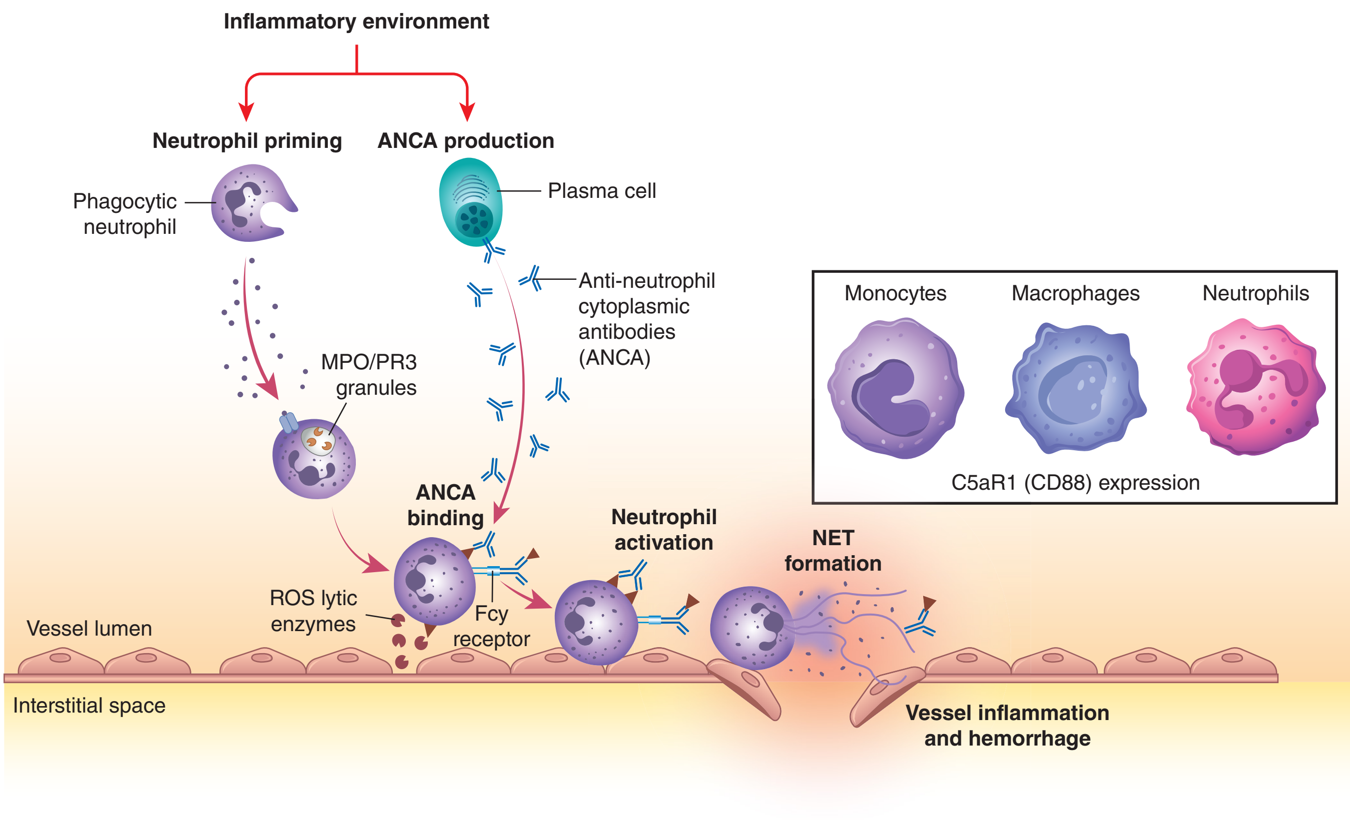

Fig. 32.2 Schematic overview of key pathogenic steps eventually leading to vasculitis. The inflammatory environment in antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis is characterized by immune cell activation, with active contribution of neutrophils, monocytes, and macrophages, which all express the C5aR1 (CD88). Alternative complement pathway activation, together with other inflammatory stimuli, leads to neutrophil priming and production of ANCA by plasma cells, among others. ANCAs bind to neutrophils, which lead to a release of reactive oxygen species and other inflammatory mediators. This leads to activation of the neutrophils and to the formation of neutrophil extracellular traps (NETs), which are key mediators of active vessel inflammation and injury. The role of NETs in antimicrobial defense is well known, but also their role in resolving neutrophil-driven inflammation in diseases such as gout. Autoantibody formation is induced by NETs in ANCA-associated vasculitis, which play a critical role in the onset of autoimmunity. Moreover, key effector cells lead to release of myeloperoxidase and proteinase 3 from granules, which are implicated in disease pathogenesis.

which acts through the CC chemokine receptor (CCR8) on mononuclear cells, correlated with the extent of cellular crescents, interstitial inflammation, and the degree of kidney function compromise and therefore also appears to be involved in the inflammatory response seen in ANCA-GN.[52] Urinary levels of T cell subsets, namely $CD3^+$ and $CD4^+$, are also increased in patients with ANCA-GN and corresponded to the crescentic class of the Berden classification (see later) when compared with the other classes.[53] In the prospective PRE-FLARED study, 102 patients ANCA-GN were recruited in remission and the predictive value or urinary $CD4^+$ T cells on renal relapses was investigated. During a median follow-up of 6 months, renal relapses occurred in 10 patients. The amount of urinary CD4+ T cells was significantly higher in patients with a subsequent relapse in comparison with those with sustained remission, with a cutoff of 490 $CD4^+$ T cells yielding a sensitivity of 60% and a specificity of 97.8%.[54]

Taken together, all these findings highlight the complexity of the pathophysiology of ANCA-GN. This growing knowledge provides a mechanistic basis for the immunosuppressive approach required to limit the extent of chemotaxis, the local inflammatory response, ongoing antibody production, and cellular innate responses that lead to kidney injury and perpetuate the inflammation (see Fig. 32.2). As our understanding grows, more targeted therapeutic approaches, with fewer risks, are becoming a reality.

DIAGNOSIS

A diagnosis of GPA or MPA is usually made by synthesis of information gathered from medical history, clinical evaluations, radiology, and eventually pathology results. Diagnostic delays are common, with a reported median time to diagnosis exceeding 6 months,[55] and contribute to the accumulation of tissue injury. A high index of suspicion is required to permit early diagnosis and treatment. Symptoms differ among patients and are related to development of acute kidney disease in cases of renal-limited forms, while most patients report nonspecific symptoms such as fever/subfebrile temperatures, weight loss, and night sweats, often in combination with organ-specific symptoms (see Tables 32.2 and 32.4). AAV is also the leading cause of "pulmonary-renal syndrome," which is characterized by the presence of acute glomerulonephritis and diffuse alveolar hemorrhage. Among autoimmune disorders leading to pulmonary-renal syndrome, AAV accounts for around 70% of cases.[56]

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

A systemic approach is required when assessing a patient with suspected AAV in order to estimate the extent and severity of organ involvement. Importantly, because PR3-ANCA or MPO-ANCA may occur in both GPA and MPA, these diseases cannot be distinguished on the basis of ANCA specificity. Practically any organ can be affected by GPA or MPA (see Tables 32.2 and 32.4). AAV often presents as multisystem disease, with a particular predilection for ear, nose, and throat (ENT), lung, and skin lesions alongside kidney disease. A high level of suspicion is raised when a patient presents with hematoproteinuria and has signs of systemic inflammation and manifestations, which are in line with a potential diagnosis of AAV. Rare manifestations can be potentially life-threatening such as aneurysms (e.g., of the renal artery),[57] can lead to sequelae significantly impacting quality of life such as orbital masses,[58] or can mimic other diagnoses such as AAV with presenting features of giant cell arteritis.[59] Patient care often involves several specialties in order to make a diagnosis. Tools such as the Birmingham Vasculitis Activity Score (BVAS)[60] are helpful to assess a patient in a systematic way and to get an impression of the severity of presentation, as well as to follow-up disease activity over time.

Patients with ANCA-GN frequently exhibit an elevation of the acute phase reactants, such as C-reactive protein (CRP), erythrocyte sedimentation rate (ESR), and platelet count during phases of active disease. Moreover, in those patients in whom the disease remains undetected for weeks to months, the eventual presentation may be with anemia of chronic disease, which might be impacted by further blood loss due to alveolar hemorrhage or gastrointestinal involvement. A systematic screen for other immunologic alterations is recommended: The presence of anti-GBM antibodies has been reported in 6% of patients with AAV,[61] a lower baseline complement C3 correlates with poor overall and kidney survival,[62] and low immunoglobulin G levels at baseline affect the risk of secondary acquired hypogammaglobulinemia after therapy initiation.[63]

A special emphasis lies on determining lung involvement in the context of AAV. The ANCA serotype-based classification allows for discrimination between patients with PR3-ANCA and MPO-ANCA disease. A study performing computed tomography (CT) scans on 129 (81 with PR3-ANCA and 48 with MPO-ANCA) patients revealed that there are distinct patterns seen in each disease. MPO-ANCA vasculitis might present with peripheral reticulation and honeycombing, characteristic of a usual interstitial pneumonitis (UIP) pattern (see Fig. 32.1A), while nodules with cavitation (see Fig. 32.1D) and central airway disease were exclusively reported in PR3-ANCA vasculitis.[20] A radiograph of the lungs is often not sensitive enough to detect some of the pulmonary lesions; thus a CT scan is the preferred method to detect overt pulmonary disease. If an infection or malignancy is suspected, the use of 18-fluorodeoxy-glucose positron emission tomography/CT might be a helpful additional diagnostic tool and might detect vasculitis involvement of other organs.[64]

Upon referral to a specialist, the median time to diagnosis in patients with generalized disease including kidney involvement was 9 days in comparison with 22 days in those with more limited disease, and the difference became more apparent when patients were seen by internal medicine physicians as opposed to ENT surgeons (6 vs. 57 days).[65] Assessment of biopsies revealed that most of the performed kidney biopsies (75%) were supportive of an AAV diagnosis, while a minority of ENT biopsies confirmed the suspected diagnosis (42%).[65]

Positivity for either PR3-ANCA or MPO-ANCA, especially at higher titers, consolidates the suspected diagnosis of GPA or MPA in most cases. Overall, only a small proportion of patients with ANCA-GN will have a negative ANCA serology (<5%), but they exhibit a typical "pauci-immune" pattern on immunofluorescence on biopsy.[66] In contrast, an expert center in China reported a high proportion of ANCA-negative "pauci-immune" glomerulonephritis, with a reported frequency of 33% in 85 patients.[67] The absence of a positive ANCA despite a consistent clinical presentation continues to pose a challenge for physicians, as ANCA-negative patients have typically been excluded from the majority of randomized controlled trials (RCTs).[10,69] Varied hypotheses have been offered to explain ANCA-negative disease. ANCA-negative vasculitis is often found when a "limited" presentation form is diagnosed (e.g., with AAV limited to the ENT tract). Of note, localized ANCA production within the affected tissue has been considered a potential explanation in these patients. In addition, other antibodies have been reported in AAV and would be of particular importance in ANCA-negative disease, but these antibodies have not been implemented in clinical practice due to lack of replication studies or limited added value in the diagnostic path to diagnose AAV. In ANCA-negative GN, infections and malignancies are present in a significant number of patients and need to be considered if there is a lack of response to therapy.[70] This also underlines that initial triggers of disease might be different from ANCA-positive cases.

HISTOLOGY

Tissue biopsy is an important component of diagnosis in AAV. In the case of a high likelihood of kidney involvement (urinalysis revealing hematoproteinuria; deterioriating kidney function), a biopsy helps substantiate the diagnosis and has moved into the center of attention because of predicting prognosis. Crescentic glomerulonephritis is a term that refers to a pattern of glomerular injury characterized by a lesion similar to the crescent of the moon. The Latin word "crescere" is used in particular for the waxing moon and by whatever coincidence, crescents in glomeruli have a huge variety of shapes similar to the different types of "lunes." From a historical perspective, the term crescentic glomerulonephritis was coined at a time when pathogenetic background of AAV was still largely unknown. It was the development of antibody staining for kidney tissue that first brought insight into the various types of crescentic glomerulonephritis because it was noticed that some appeared in combination with a particular staining pattern denoting the presence or absence of immune complexes or other deposits (see Table 32.1). Today, the most common forms of crescentic glomerulonephritis are pauci-immune ANCA-GN and anti-GBM nephritis, which are described in detail in this chapter. However, crescents can occur in many more kidney diseases such as IgA nephropathy, lupus nephritis, infectious GN, C3 glomerulonephritis, and even, more unexpectedly, in membranous nephropathy

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

or drug-induced glomerulonephritis, which are discussed further elsewhere in this book (see Chapters 30-34 and 36).

Pauci-immune ANCA-GN is a form of crescentic glomerulonephritis characterized by a *paucity* of both immunoglobulins and components of the complement system in glomeruli as tested by antibody staining (immunofluorescence). The term was first used in 1989 by Jennette, Wilkman, and Falk.[71] This form of crescentic glomerulonephritis, with its lack of immune-complex deposition, is consistent with the pathogenesis of ANCA-GN based on a type II hypersensitivity reaction in which neutrophils are activated by autoantibodies in the circulation, ultimately leading to vasculitic lesions in the absence of immune-complex deposition. Pauci-immune GN was originally defined as a staining pattern by immunofluorescence in which there was less than 2+ staining for immunoglobulins on a scale from 0 to 4, as well as for complement components.[72] More recently, however, exclusive and strong staining for C3 in patients with ANCA-GN has been reported to occur in up to 50% of patients.[73] Scaglioni and colleagues[74] were the first to question whether ANCA-GN is always pauci-immune, with results demonstrating substantial deposition of IgG and/or C3 in the capillary walls and/or mesangial areas of around 25% of a study group of 48 patients. Another study of 187 kidney biopsies from patients with ANCA-GN found positivity for C3c in 42% and showed their presence by electron microscopy (EM) in both mesangial areas and along the GBM.[75] C3 can either occur uniquely or in combination with IgG in ANCA-GN and may be associated with a slight difference in clinical presentation. In an early study, the presence of immunoglobulins in ANCA-GN has been associated with higher levels of proteinuria,[76] whereas another study found that glomerular deposition of C3 in ANCA-GN was associated with worse renal and overall survival.[77] Strongly positive C3 staining (Fig. 32.3H) in glomeruli occurs in patients with both PR3-ANCA and MPO-ANCA. In contrast, a retrospective multicenter study of 154 patients concluded that the lack of clear predictive value of C3 serum levels and the finding of similar kidney outcomes irrespective of C3 deposition on biopsy precludes the use of C3 levels or staining as biomarkers of AAV outcomes.[78] However, in view of new therapies with complement inhibitors, it would be of interest to study whether these drugs have different outcomes in patients with or without signs of complement activation on biopsy—these data are not yet available.

Necrotizing crescentic glomerulonephritis with a pauci-immune pattern is the most characteristic finding of ANCA-GN. Glomerular fibrinoid necrosis (see Fig. 32.3A) is representative of small vessel vasculitis at the level of the glomerular capillary loops, which leads to crescent formation in the Bowman space (see Fig. 32.3B and 32.3C). There may be marginally different combinations of lesions in patients with either PR3-ANCA or MPO-ANCA, where the former tend to exhibit more acute lesions and the latter more chronic lesions,[79] but no lesions are uniquely present in either serologic group. Renal arteritis (see Fig. 32.3D) is encountered in 13.5% of patients with ANCA-GN, and these patients are older, have a more pronounced inflammatory infiltrate, and have a worse kidney prognosis.[80] Although a granulomatous reaction is still considered to be the essential inflammatory reaction in AAV, granulomas are almost never encountered in kidney biopsies of patients with AAV[79] (see Fig. 32.3E and 3F). The growing awareness of AAV among clinicians coincides with its earlier detection in recent years, which may be responsible for this. However, rupture of the Bowman capsule that may precede the granulomatous reaction around a glomerulus has recently gained interest.[81] Necrotizing vasculitis of medium and large sized arteries is also a rare finding,[80] present in no more than 15% of kidney biopsies in large studies. However, this is most likely related to the "skiplike" nature of these lesions leading to sampling error. Most kidney biopsies in ANCA-GN show a fair amount of acute interstitial inflammation, and some studies reported a relation between tubulitis (see Fig. 32.3G) caused by these inflammatory infiltrates and renal outcome.[82] Fig. 32.3 shows the most characteristic lesions of ANCA-GN.

By light microscopy, crescents in many different diseases appear to be the same and represent a common histologic endpoint. A crescent may develop where there are breaks in the glomerular capillary basement membrane,[83] assuming that whatever caused this break is irrelevant. Breaks in the glomerular basement membrane lead to the entry of inflammatory mediators into the Bowman space that initiate the proliferation of cells lining the Bowman capsule. The proliferative nature of this process is reflected in the cytonuclear characteristics of the cells including variability in nuclear size, slightly atypical features such a rather strong chromatin density and prominent nucleoli, and an occasional mitotic figure indicative of the rapid multiplication of the epithelial cells. Besides epithelial cells, crescents have many other components, most importantly a large variety of inflammatory cells such as lymphocytes, monocytes, macrophages, and granulocytes. A number of studies have shown a relationship between urinary soluble CD163 and the presence of crescents on the kidney biopsy, suggesting that tissue changes may be reflected in the urine.[47]

A long-standing topic of discussion has been the role and involvement of podocytes in crescent formation. Whereas epithelial cells lining the Bowman capsule are referred to as the parietal epithelial cells of the Bowman space, the podocytes are referred to as the visceral epithelial cells and these two cell types develop from the same lineage.[84] Both cell types seem to be involved in crescentic lesions. However, there may be a difference in their contribution where, for instance, parietal epithelial cells are more prominently present in the pseudocrescents of collapsing focal segmental glomerulosclerosis (FSGS) and visceral epithelial cells are more dominant in the crescentic lesions of pauci-immune ANCA-GN. The role of podocytes in clinical manifestations of patients with ANCA-GN has not been thoroughly investigated; one study demonstrated that podocyte loss in ANCA-GN is associated with proteinuria.[85] Interestingly, a morphometric signature of podocyte depletion in ANCA-GN was found by a deep learning-based approach for antigen-specific cellular morphometrics, showing potential for risk stratification.[86] For more discussion on glomerular cell biology and podocytopathies, see Chapter 4.

Kidney pathologists use varied terminologies to describe the extent of crescent formation and its development from an acute to a chronic lesion. For the latter, there is the distinction between cellular, fibrocellular, and fibrous crescents. To describe the extensiveness in a single glomerulus, the terms *segmental* (Fig. 32.4A) and *circumferential* (Fig. 32.4C–4D)

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

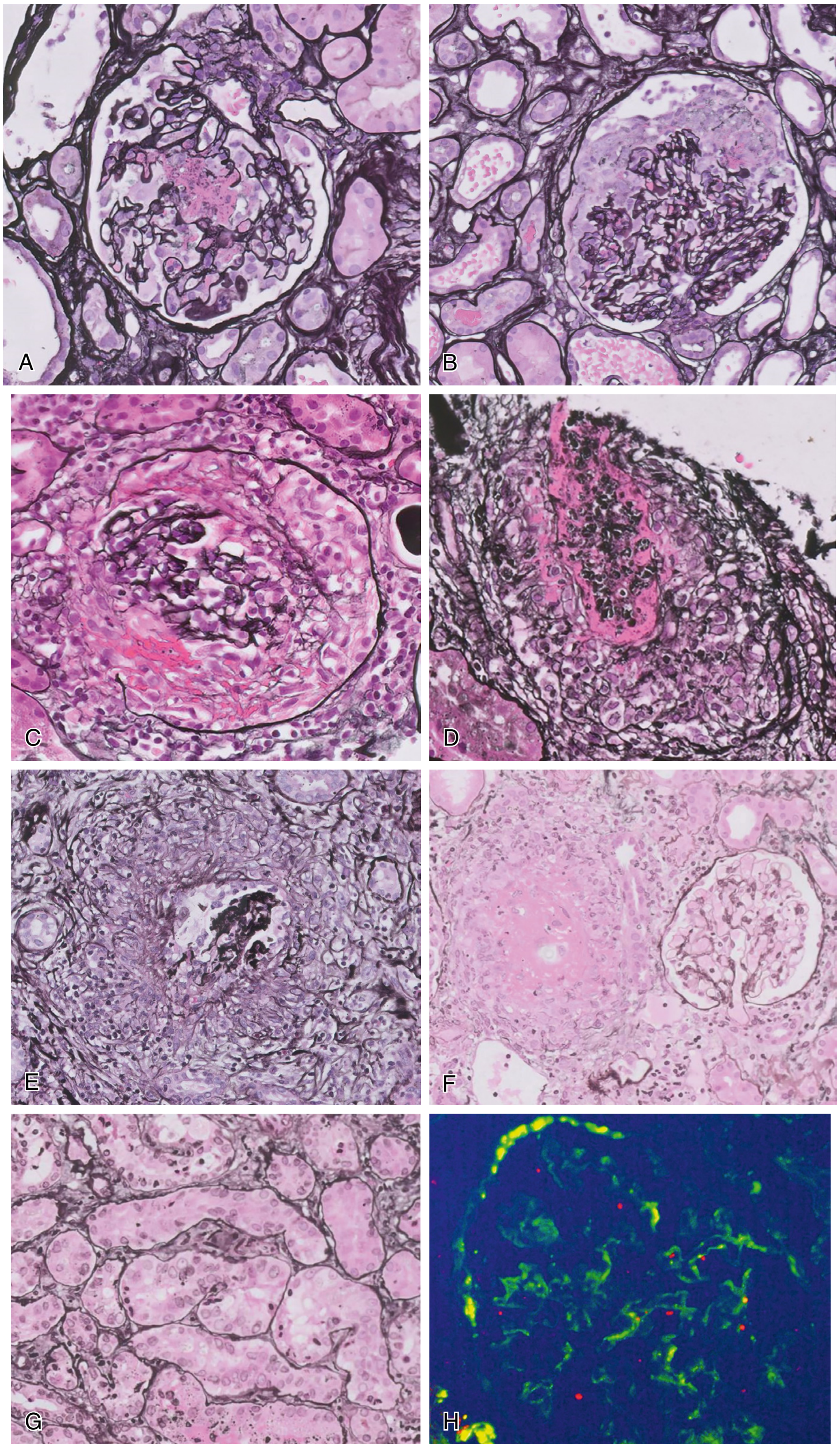

Fig. 32.3 Characteristic lesions in antineutrophil cytoplasmic antibody glomerulo nephritis. (A) Fibrinoid necrosis representative of vasculitis at the level of capillary loops. (B) Crescent formation: cellular crescent around area with breaks in the GBM and small focus of fibrinoid necrosis. (C) Circumferential cellular crescent with a break in the Bowman capsule: inflammation extends into the interstitial area. (D) Necrotizing vasculitis of a small artery in kidney biopsy specimen. (E) Perigranulomatous reaction around an almost destructed glomerulus. Granulomatous reaction with palisading of histiocytes. (F) Interstitial granuloma with necrosis in its center; could represent total destruction of a glomerulus or other no longer recognizable structure. (G) Tubulitis: cytotoxic T-lymphocytes invade the tubular epithelium. (H) Immunofluorescence for C3 in a glomerulus: positivity along the GBM and in mesangial areas.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

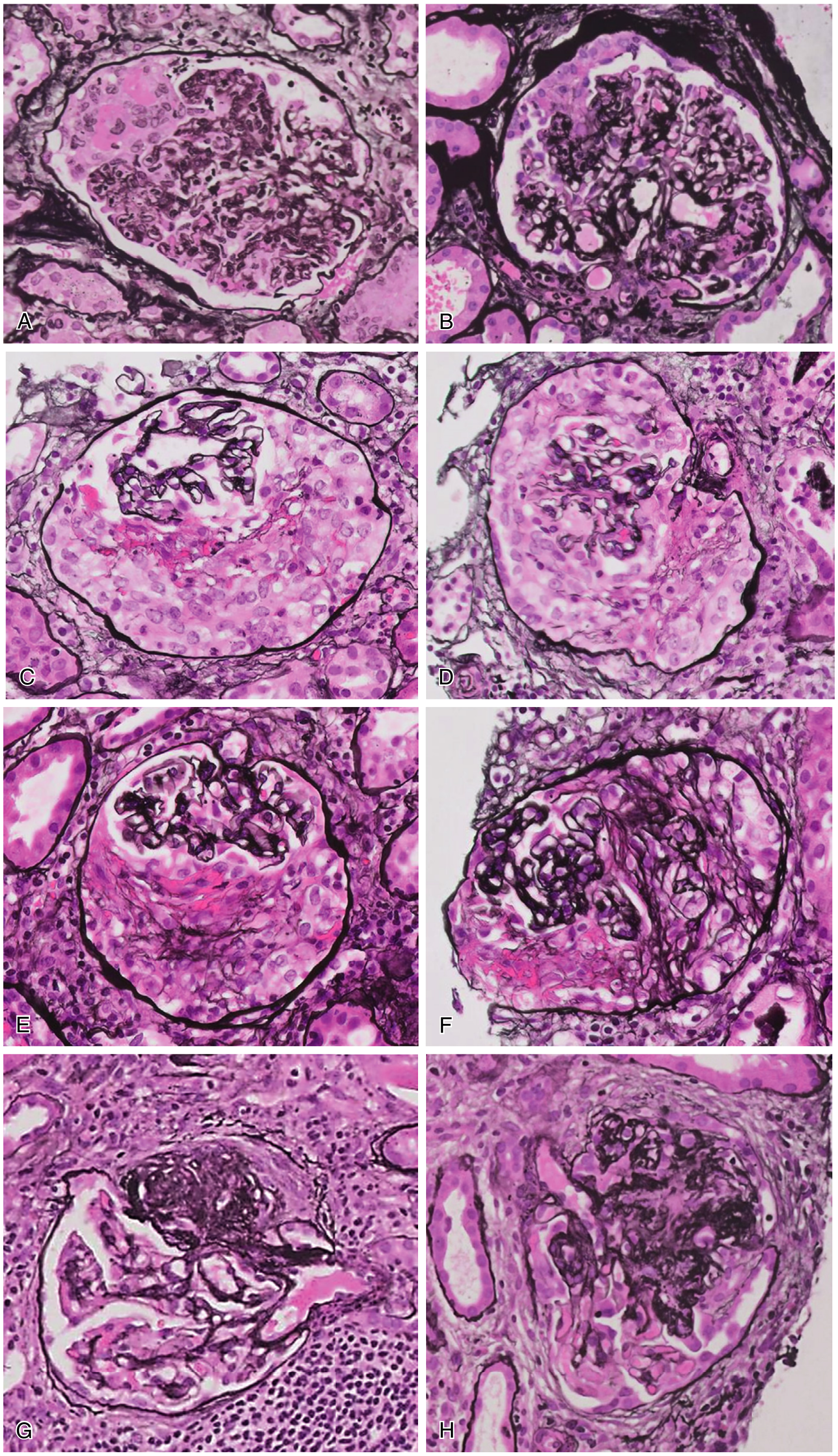

Fig. 32.4 Stages of a crescent. (A) Segmental cellular crescent with fibrinoid necrosis. (B) Cellular crescent showing the contribution of visceral and parietal epithelial cells. (C) Almost circumferential cellular crescent with streaks of fibrin. (D) Circumferential cellular crescent: a few collagen fibers begin to appear. (E) Fibrocellular crescent. (F) Extension of collagen fibers moving into fibrous crescent. (G) Segmental lesion probably resulting from crescent resembling FSGS. (H) Continuation of chronic lesions leads to an almost globally sclerotic glomerulus.

are used. To give an indication of the severity of crescentic glomerulonephritis, the terms *focal* and *diffuse* are used: focal if crescents are present in less than half of all glomeruli in a biopsy, diffuse if they are present in more than half of all glomeruli in a biopsy (see Fig. 32.4). For detailed descriptions of the composition of crescents, slightly different definitions developed over time for different diseases. Kidney pathology working groups are currently working on the harmonization

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

of these definitions.[87,88] Having the same definitions for the descriptors of the crescentic lesions in all kidney diseases will be helpful for standardization purposes. However, this unification should not withhold us from investigating whether similar patterns of crescent formation in different diseases have the same implications, for instance, with regards to outcome parameters and treatment decisions. We know that crescents have a variety of pathogenetic routes. In day-to-day diagnostics, the immunofluorescent or immunohistochemical procedures that clinical pathologists use to determine whether the crescents are representative of kidney involvement in AAV, systemic lupus erythematosus, and many other diseases reflect the enormous variability of conditions under which crescents may develop on a clinical basis alone. Therefore we cannot exclude that there is also variability in the significance of crescents for different diseases, but little is known about this. That the presence of 100% crescents in anti-GBM disease is regarded as a sign of irreversible damage not responsive to treatment[89] is an example of how the crescentic pattern of disease is used for management purposes—but pertaining to anti-GBM disease only. The rule of thumb of more than 50% crescents in a kidney biopsy being representative of clinical RPGN is another example, perchance related to another 50% rule in the histopathologic classification for ANCA-GN commonly referred to as the Berden classification.

In order to classify patterns of disease in pauci-immune ANCA-GN that have a relation to outcome, the Berden classification[90] established 4 histologic classes based on 50% or more glomeruli in a biopsy having no abnormalities (focal class), having cellular crescents (crescentic class), having global sclerosis (sclerotic class), or otherwise put into a mixed category (mixed class). By combining clinical and histologic parameters, a renal risk score for kidney function outcome was later developed by Brix and colleagues,[91] the so called ANCA Renal Risk Score (ARRS, also referred to as the Brix score), recently improved as the kidney risk score (AKRiS)[92] and now containing four risk classes in order to optimize clinicopathologic prognostication for clinical practice and trials. With the rapid development of new drugs, these two tools need to be kept up to date in order to enhance their prognostic value. Other scores used to assess biopsies of patients with ANCA-GN are the Mayo Clinic Chronicity Score (MCCS)[93] and Percentage of ANCA Crescentic Score (PACS,[94] Table 32.5). More and more kidney biopsies from patients with AAV tend to have a focal class pattern of disease in comparison with a decade ago. In kidney biopsies that lack typical glomerular lesions, it is advised to work up all of the tissue: Due to their focal distribution, lesions may not be apparent at each level.[95] In fact, case reports of acute interstitial nephritis as the only lesion in a patient with AAV (i.e., in the absence of crescent formation) could be examples of patients in whom the number of glomeruli with crescent formation is below the level of detection in a biopsy with a moderate amount of glomeruli.

VASCULITIS MIMICKERS

Several scenarios exist where detection of PR3-or MPO-ANCA is misleading. A single-center study from France analyzed 288 ANCA-positive patients, of whom 49 had a final diagnosis of either GPA or MPA. Titers >3.25-fold over the reference limit of a positive test had a sensitivity of 94% and a specificity of 73% to discriminate AAV from mimickers. The other 239 patients either had another autoimmune disorder (total n = 99; inflammatory bowel disease, n = 16; systemic lupus erythematosus, n = 14; non-AAV vasculitides, n = 11) or in the context of other diseases (total n = 140; infections, n = 38; myocardial infarction/stroke, n = 25; hematologic disorders, n = 13),[96] highlighting that ANCA-positivity is frequently encountered in varied scenarios. Similar findings were reported previously in patients with MPO-ANCA–positivity. Importantly, almost 50% of these patients exhibit signs of kidney disease, either reduced estimated glomerular filtration rate (below 60 mL/min/1.73 m^2), overt proteinuria, or microscopic hematuria.[97] Patients with AAV on average have a higher CRP in comparison with those who present with other diagnoses but are tested ANCA–positive.[96] A higher CRP value is certainly encountered in cases with ANCA–positivity and underlying infections. Infective endocarditis has been associated with PR3-ANCA–positivity, as approximately 20% are ANCA–positive. These patients more frequently have a longer period of infection and present with purpura, macrohematuria, proteinuria, and acute kidney injury. Some of these patients have a “pauci-immune” pattern when kidney biopsies are performed.[98] In cases with suspected systemic infections and ANCA positivity, determining procalcitonin levels might be helpful to ease distinguishing between infections and AAV.[99] Various viruses, bacteria, fungi, and protozoa have been associated with the formation of ANCA.[100]

An analysis of the World Health Organization pharmacovigilance database (VigiBase) identified 483 deduplicated case records of drug-induced AAV. These patients were on average 62 years, were more frequently female (71%), and there was a delay of 9 months between introduction of the new medication and onset of disease. The three drugs with the highest disproportionate reporting and the highest number of reported cases were hydralazine as an antihypertensive agent and propylthiouracil and thiamazole as antithyroid drugs.[101] Among antithyroid drugs, propylthiouracil has a greater potential to induce drug-induced vasculitis.[102] Drug-induced vasculitis usually presents with MPO-ANCA or double positivity, defined as presence of MPO-ANCA and PR3-ANCA at the same time. The detection of other antibodies, such as antinuclear antibodies, is also common in drug-induced vasculitis. Management of such cases include immediate drug cessation and introduction of immunosuppression if patients present with severe disease (e.g., kidney involvement). Maintenance of remission is not required in most cases when the offending substance is withdrawn, and rechallenging of the patient with the drug implicated in inducing the vasculitis should be avoided.[103] Other medications include allopurinol, minocycline, penicillamine, rifampicin, and sulfasalazine, among others. However, causal associations between these drugs and AAV are not well established.

Childhood AAV is rare and has a female preponderance.[104] A rare autosomal dominant genetic disease, “Copa syndrome,” can mimic severe AAV including kidney disease and alveolar hemorrhage and typically occurs early in life with disease onset usually younger than 5 years of age and has a female predominance.[105]

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.5 Different Histopathologic Scores have been Reported to Score Biopsies of Patients with ANCA-Glomerulonephritis

	Berden[90]	AKRiS[92]	MCCS[93]	PACS[94]
Glomerular lesions	≥50% normal glomeruli (focal) ≥50% cellular crescents (crescentic) ≥50% glomerulosclerosis (sclerotic) All others (mixed)	Normal glomeruli (%) >25 (N0), 10-25 (N1), <10 (N2)	Glomerulosclerosis <10%; 10%-25%; 26%-50%; >50%	% of the following lesions: Cellular crescent Fibrocellular crescent Fibrous crescent Sclerosis (43% predictive of KF)
Tubulointerstitial lesions		IFTA (%) <25 (T0), ≥25 (T1)	Interstitial fibrosis Tubular atrophy (both <10%; 10%-25%; 26%-50%; >50%)	
Vascular lesions			Atherosclerosis None/mild Moderate/severe	
Clinical characteristics		Creatinine (μmol/L) <250 (K0), 250-450 (K1), >450 (K2)		

The ANCA Kidney Risk Score (AKRiS) combines pathology and baseline characteristics and has the best ability to predict kidney failure. The above-mentioned scores have been independently validated, especially the Berden histopathologic classification.
IFTA, Interstitial fibrosis, tubular atrophy; *KF,* kidney failure.

COCAINE AND LEVAMISOLE-ASSOCIATED ANCA DISEASE

One particular observation is noteworthy. The use of illicit substances including cocaine consumption varies among countries, rural and urban areas, and patient population but is generally increasing. Around 70 percent of illicit cocaine in the United States is contaminated or "cut" with levamisole, an anthelminthic drug with immunomodulatory effects, which can induce AAV with a predominant widespread cutaneous involvement. These patients are generally younger and present with double positivity (i.e. both c-ANCA and p-ANCA) and other autoantibodies. The largest case series reported ANCA-GN in around half of the affected patients.[106] Cocaine per se can also cause AAV, and these patients frequently present with severe localized ENT involvement, often causing septal perforation. Kidney disease is infrequent in such patients; most will present with PR3-ANCA. In younger patients, testing the urine for cocaine and levamisole (if available) is advised. Therapy comprises avoidance of exposure, and many will require immunosuppressive therapies. Of note, therapy response is limited if patients continue to use cocaine.[107]

THERAPY

Therapeutic strategies have historically been divided into those used in the induction of remission (within the first 3–6 months of active disease) and those used to maintain remission (subsequent therapies). The aim of the induction phase of therapy is to rapidly reduce inflammation and achieve remission, which is usually defined as a BVAS of zero (according to, e.g., version 3)[60] and has been used in most clinical trials. A change in BVAS has been proven to be repeatable, reproducible, and sensitive.[60] The renal items of BVAS are often semiquantitative, and a maximum of 12 points is reached easily (hematoproteinuria and a form of creatinine increase) but does not always reflect the severity of kidney function impairment; that means a patient with a creatinine of 1.7 mg/dL or 8.1 mg/dL and concomitant hematoproteinuria would have the same score. An overview of key clinical trials dictating current therapeutic strategies is highlighted in Fig. 32.5.

INDUCTION OF REMISSION

Steroid Reduction and Steroid-Sparing Strategies

Glucocorticoids are a main pillar of therapy in ANCA-GN and are able to rapidly reduce inflammation. Thus glucocorticoids are a mainstay of therapy in AAV. However, glucocorticoid toxicity contributes to morbidity and mortality of patients with AAV by increasing the risk of infections, and among other side effects, it has an impact on cardiovascular risk factors such as blood pressure and lipids.[108] No clinical trial has investigated whether initial therapy with high-dose methylprednisolone pulses has beneficial effects on kidney function recovery. A retrospective study of 114 patients from 5 expert centers with either a serum creatinine >500 μmol/L (5.7 mg/dL) or dialysis dependency at enrollment found that patients who received intravenous pulse methylprednisolone (followed by oral glucocorticoids) did not have higher rates of recovery of kidney function (57.7% in the methylprednisolone arm vs. 66.1% in the oral prednisolone alone arm), which was defined as either independence from kidney replacement therapy in those who presented with dialysis dependence or an improvement of kidney function in those not requiring dialysis at baseline.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

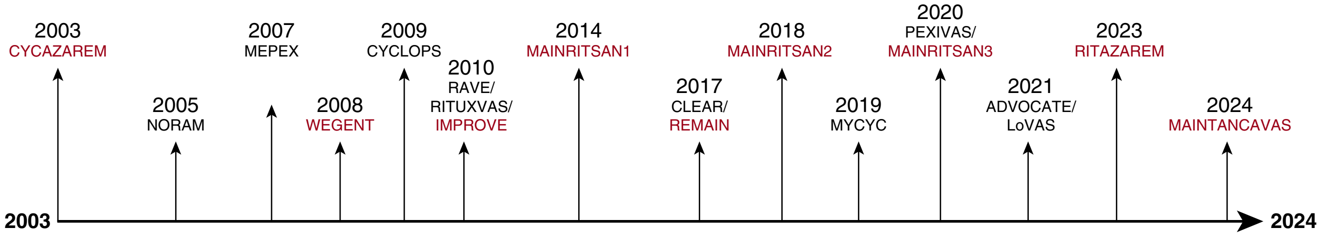

Fig. 32.5 Timeline of relevant therapeutic developments in antineutrophil cytoplasmic antibody–associated vasculitis over the past 2 decades. Trials testing new strategies to induce remission are given in black, while trials testing therapeutic options to maintain remission are highlighted in red.

However, the patients receiving methylprednisolone pulses experienced more frequent serious infections by 3 months (36.5% vs. 19.4%), and more developed diabetes by 12 months (26.9% vs. 6.5%) than those who had received oral prednisolone alone.[109] The total oral prednisolone dose was 4 g in the methylprednisolone group versus 6.8 g in the oral prednisolone group, suggesting that the initial high-dose therapy increased the risk of complications, and a special focus on prevention of infections is required when methylprednisolone pulses are used.

The first clinical trial to investigate substantial reduction of glucocorticoid exposure was the RAVE trial, which advised investigators to stop prednisone after 5.5 months, and this was part of the primary endpoint. The cumulative glucocorticoid dose by 6 months was 4.0 g in the rituximab arm and 4.4 g in the cyclophosphamide/azathioprine arm.[22] Two further clinical trials have studied the effect of reduced doses of glucocorticoids on either the composite of kidney failure and death (PEXIVAS)[23] or remission (LoVAS).[110] In the PEXIVAS trial, participants receiving the reduced dose of glucocorticoids had an exposure that was <60% at 6 months of that in the standard-dose group (for regimen comparison, see Table 32.3). In terms of efficacy, the reduced-dose regimen was equivalent to the standard dose in terms of kidney failure and mortality, but the frequency of serious infections was reduced, as 96 patients (27.2%) in the reduced-dose arm and 116 patients (33.0%) in the standard-dose arm developed at least one serious infection (HR 0.69, 95% CI 0.52–0.93).[23] Of note, most patients in the PEXIVAS trial received cyclophosphamide-based induction therapy, so information on rituximab induction plus reduced-dose glucocorticoids remains limited. A retrospective real-world study from France found that a combined endpoint, including progression of disease, relapse rate, kidney failure, and death, was reached more frequently in patients receiving a reduced-dose PEXIVAS regimen; however, important information on achieving the endpoints (definition of progression) and the highly imbalanced baseline characteristics are significant issues of this study.[111]

The LoVAS trial, conducted in Japan, randomized patients to a reduced-dose or high-dose glucocorticoid regimen in addition to rituximab therapy. Most patients had MPO-ANCA vasculitis (85.8%), and the eGFR at baseline was relatively well preserved in both arms (52.0 vs. 55.3 mL/min/1.73 m^2). The cumulative dose of glucocorticoids received in the reduced-dose group was 1.3 g, which was 68.3% lower compared with the high-dose group (cumulative exposure of 4.2 g) (see Table 32.6 for glucocorticoid dosing). Remission rates were equivalent between the two glucocorticoid groups, while less treatment toxicity and fewer serious infections (5 vs. 13 patients) were observed with the reduced-dose glucocorticoids ($P = 0.04$).[110] Longer-term follow-up showed that the reduced-dose regimen was associated with fewer serious adverse events, while the relapse rate did not differ between groups.[112] Lower doses of glucocorticoids with exposure limited to 2 weeks only have been reported in observational studies using a combination therapy of cyclophosphamide and rituximab as induction of remission treatment (see later).[113] Although clinical trials have provided evidence that reduction of cumulative doses and earlier withdrawal (within the first 6 months) of glucocorticoids is noninferior in terms of achieving remission and future relapses, the duration of glucocorticoid therapy is still a controversial topic.

In patients with ANCA-GN, the reduced-dose PEXIVAS glucocorticoid regimen is nowadays standard and recommended by all major recommendations/guidelines.[114] A steroid-free remission maintenance (i.e., withdrawal of glucocorticoids before 6 months) is possible and safe, especially when rituximab as maintenance agent is used, and is preferred by the authors of this chapter.

Complement Inhibition with Avacopan

Another option to reduce cumulative glucocorticoid exposure is the addition of avacopan, an oral C5aR1 inhibitor. It has been tested in animal models of MPO-ANCA vasculitis[43] and was tested in two smaller phase 2 trials: CLEAR,[51] a dose-finding study conducted in Europe, and CLASSIC,[115] a safety study conducted in the United States. The phase 3 trial, ADVOCATE, was an international study and has assigned 330 patients (268; 81.2% with ANCA-GN) to either receive avacopan (30 mg twice daily, ± a short course of prednisone for a maximum of 4 weeks, maximum initial dose 20 mg) or prednisone, which was tapered off over 20 weeks (see Table 32.6). All patients were either receiving rituximab (4-times 375 mg/m^2, a week apart each; used in 214 patients (64.9%)) or cyclophosphamide (either intravenous as used in 102 patients [30.9%] or orally in 14 [4.2%]). An eGFR of <15 mL/min/1.73 m^2 was an exclusion criterion of ADVOCATE.

Two endpoints were tested: 1. BVAS of 0 at week 26 and no glucocorticoid therapy in the previous 4 weeks and 2. sustained remission, defined as remission at weeks 26 and 52 and no receipt of glucocorticoids for 4 weeks. The first primary endpoint was observed in 72.3% and 70.1% of patients assigned to avacopan and prednisone, respectively (noninferiority), while the second primary endpoint

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.6 Different Steroid Tapering Regimen According to PEXIVAS, LoVAS, and ADVOCATE Trials

Trial	PEXIVAS			LoVAS			ADVOCATE	
Regimen	Reduced dose	Standard dose		Low dose	High dose			
Weeks	<50/50-75/>75 kg	<50/50-75/>75 kg	Weeks			Weeks	>55 kg	<55 kg
1-2	25/30/40 mg	50/60/75 mg	1-2	0.5 mg/kg	1.0 mg/kg	1	60 mg	45 mg
3-4	20/25/30 mg	40/50/60 mg	3-4	0.25 mg/kg	0.8 mg/kg	2	45 mg	45 mg
5-6	15/20/25 mg	30/40/50 mg	5-6	7.5 mg	0.7 mg/kg	3	30 mg	30 mg
7-8	12/15/20 mg	25/30/40 mg	7-8	5 mg	0.5 mg/kg	4-6	25 mg	25 mg
9-10	10/12/15 mg	20/25/30 mg	9-10	4 mg	0.4 mg/kg	7-8	20 mg	20 mg
11-12	7/10/12 mg	15/20/25 mg	11-12	3 mg	0.35 mg/kg	9-10	15 mg	15 mg
13-14	6/7/10 mg	12/15/20 mg	13-16	2 mg	15 mg	11-14	10 mg	10 mg
15-16	5/5/7 mg	10/10/15 mg	17-20	1 mg	12.5 mg	15-20	5 mg	5 mg
17-18	5/5/7 mg	10/10/15 mg	21-24	0 mg	10 mg	≥21	0 mg	0 mg
19-20	5/5/5 mg	7/7/10 mg						
21-22	5/5/5 mg	7/7/7 mg						
23-52	5 mg	5 mg						
>52	Investigators' local practice	Investigators' local practice						

The PEXIVAS trial did not mandate steroid withdrawal, and further reduction steps beyond 52 weeks of follow-up were based on the investigators' local practice. The LoVAS trial assessed its primary endpoint at 6 months of follow-up, while steroid withdrawal formed the basis of both primary endpoints of ADVOCATE.

was observed in 65.7% and 54.9%, respectively, showing superiority of an avacopan-based therapy over prednisone. Of note, relapses were less common in the avacopan arm (HR 0.46; 95% CI 0.25–0.84). Glucocorticoid toxicity was significantly reduced by the addition of avacopan. In 265 of 268 patients with ANCA-GN, eGFR at baseline was available. Kidney function improved in those with ANCA-GN over time in the avacopan and the prednisone arm but was greater in the avacopan arm with an eGFR increase of 7.3 in contrast to 4.1 mL/min/1.73 m^2 in the prednisone arm. eGFR increase was recorded from a baseline eGFR of 44.6 and 45.6 mL/min/1.73 m^2 in the treatment arms.[116] Investigators focused on the 50 patients (of 330) who were recruited with an eGFR of ≤20 mL/min/1.73 m^2 and observed an eGFR recovery of +16.1 and 7.7 mL/min/1.73 m^2 in the avacopan and prednisone groups, respectively.[117]

Avacopan is the first eGFR-sparing therapy in ANCA-GN and might be of particular value for those patients presenting with severely impaired kidney function at baseline. Of note, those patients with an eGFR <15 mL/min/1.73 m^2 were excluded from the trial and there is no information about avacopan efficacy in patients with dialysis dependency. The substance also allows for substantial reduction of cumulative glucocorticoid exposure and thus is of relevance for those facing glucocorticoid-related side effects and toxicity. Other inhibitors of the alternative complement pathway, such as the factor B inhibitor iptacopan, are currently tested in clinical trials of AAV (NCT06388941). However, a more proximal inhibition of the complement system does involve the addition of prophylaxis against encapsulated bacteria such as meningococcal infections.[118]

Rituximab, Cyclophosphamide, or Combination Therapy

The 2022 update of the EULAR recommendations for the management of AAV recommend that either rituximab or cyclophosphamide is used to treat organ-threatening or life-threatening disease.[119] This recommendation is based on the publication of the RAVE trial, which randomized patients with new or relapsing disease to either rituximab (4 × 375 mg/m^2, a week apart; n = 99) or oral cyclophosphamide (2 mg/kg body weight, with adjustments; n = 98), followed by azathioprine (2 mg/kg body weight) as an agent to maintain remission. At 6 months, 64% in the rituximab arm as opposed to 53% in the control arm reached the primary endpoint, defined as BVAS/WG of 0 and a prednisone dose of 0 mg whereby rituximab achieved noninferiority.[120] At 18 months, only 39% of those who received rituximab and 33% of controls remained in remission. In the rituximab arm, patients did not receive maintenance therapy, while the control group was maintained on azathioprine until the end of the trial. In trial participants with relapsing disease, the rates of remission were significantly different at 6 and 12 months (67% and 49% vs. 42% and 24%; for rituximab vs. cyclophosphamide/azathioprine), but a borderline significance was reported at 18 months (37% vs. 20%; for rituximab vs. cyclophosphamide/azathioprine). Relapses observed during the RAVE trial were mainly driven by individuals who had a historic or present positivity for PR3-ANCA.[22] In patients with relapses (n = 81), a significantly higher proportion remained in remission when assigned to rituximab.[121] The RAVE trial excluded patients with a serum creatinine ≥4.0 mg/dL,[120] leaving a knowledge gap from RCTs on rituximab use in those presenting with severe kidney disease. The PEXIVAS trial included 704 patients

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

with a median serum creatinine of 3.8 mg/dL, where 109 (15.5%) trial participants received rituximab alongside plasma exchange (PLEX; $n = 55$) or no-PLEX ($n = 54$). Fewer patients reached the composite of kidney failure and death at a median follow-up of 2.9 years when PLEX was added to rituximab and glucocorticoids, which was comparable with the overall outcome of the trial and therefore nonsignificant (see later for more details).[23] Subanalyses focusing on kidney outcomes (kidney failure within the first year, increases in eGFR) of patients in PEXIVAS are ongoing and will shed more light on rituximab' efficacy in patients with severe kidney disease because currently available data are limited to mainly retrospective observational data. The utility of PLEX itself is discussed in more detail later.

Combination of rituximab and cyclophosphamide has become a popular strategy for induction of remission in severe ANCA-GN. A combination of rituximab (4-times 375 mg/m^2) and cyclophosphamide (2 scheduled infusions, 15 mg/kg body weight, adjusted) was tested in the RITUXVAS trial and found to be noninferior to 3 to 6 months of intravenous cyclophosphamide are used according to the CYCLOPS trial (see Table 32.7), followed by azathioprine as maintenance agent. This trial was small (33 and 11 patients) but found a nonsignificant increase in eGFR of 19 mL/min/1.73 m^2 in the combination group compared with 15 mL/min/1.73 m^2 in the cyclophosphamide-azathioprine arm.[122] Further evidence that combined therapy with rituximab and cyclophosphamide is effective stems from observational studies.[113,123] However, the retrospective nature of these studies and lack of a comparator in some studies are limitations of these studies.

Variable approaches to concomitant glucocorticoid use and the use of cyclophosphamide/rituximab and PLEX have been employed by different groups in prospective RCTs as summarized in Table 32.7. PLEX was used in all patients in one study,[124] while it was added to the therapeutic regimen in some cases in the other studies. Overall, it seems that a more intense initial immunosuppressive regimen may allow significant reduction in glucocorticoid exposure, and the study by Pepper and colleagues[125] limited oral glucocorticoid exposure to a maximum of 2 weeks, thus enabling steroid minimization in patients presenting with severe AAV. Cortazar and colleagues[123] used oral cyclophosphamide (adjusted for kidney function) together with two initial doses of rituximab and glucocorticoids, and a third of the patients also underwent PLEX. Remission was achieved by 84% at 5 months, and glucocorticoids were discontinued at 8 months. In patients presenting with the most severe kidney disease ($n = 75$), the authors reported a significantly greater increase in eGFR in patients with PR3-ANCA vasculitis in comparison with MPO-ANCA–positive ones (+16 vs. +5.6 mL/min/1.73 m^2). Relapses were only observed in patients with PR3-ANCA vasculitis (6/20 versus 0/55 patients),[123] highlighting that the potential for recovery of kidney function is better but relapse risk is higher in patients with PR3-ANCA vasculitis.

KDIGO and EULAR both acknowledge the paucity of data from high-quality clinical trials to use rituximab in patients with severe kidney disease. In such scenarios, the use of low-dose cyclophosphamide (e.g., two initial pulses) alongside rituximab is an attractive opportunity and has been tested in the RITUXVAS trial. The combination has been reported in several observational studies (see Table 32.7). In patients with a serum creatinine <4 mg/dL, rituximab (without cyclophosphamide) has become the standard in most centers.

The Role of Plasma Exchange

Hardly any topic in nephrology is as controversial as the use of PLEX in AAV. A survey capturing the opinion from key researchers working in the field of AAV revealed that 39% still use PLEX as part of their therapy to induce remission, and this response was more common among nephrologists and participants who are managing a higher AAV patient volume. PLEX is used more often in Europe than in North America and Asia.[127] Two key trials form the basis for our current understanding of the efficacy of PLEX in AAV, with substantial differences in the concomitant therapies: the MEPEX trial published in 2007[128] and the PEXIVAS trial in 2020.[23] MEPEX randomized patients with a serum creatinine >500 µmol/L (5.7 mg/dL) to receive a series of 7 exchanges within 14 days or 3 doses of 1000 mg/day intravenous methylprednisolone on top of oral cyclophosphamide at a dose of 2.5 mg/kg/day (reduced for age >60 years), which was subsequently reduced at 3 months, and oral glucocorticoids. At 1 year, patient survival and severe adverse event rates were 76% (51 out of 67) and 48% (32 out of 67) in the intravenous methylprednisolone group and 73% (51 out of 70) and 50% (35 out of 70) in the PLEX group, respectively. The percentage of patients experiencing recovery of kidney function was higher in the PLEX group. Specifically, 10 (19%) out of 51 survivors in the PLEX arm remained dialysis dependent at 12 months in comparison with 22 (43%) out of 51 survivors in the methylprednisolone group. Those who recovered kidney function had an improvement in serum creatinine from 718 µmol/L (8.2 mg/dL) to 198 µmol/L (2.3 mg/dL) in the methylprednisolone arm and from 754 µmol/L (8.6 mg/dL) to 199 µmol/L (2.3 mg/dL) in the PLEX arm. In both treatment arms the risk of severe/life-threatening leukopenia and infections was high,[128] presumably based on the high doses of concomitant therapies. The PEXIVAS trial wanted to expand the indication of PLEX and recruited patients with an eGFR <50 mL/min/1.73 m^2, with a median serum creatinine of 327 µmol/L (3.7 mg/dL) in the PLEX and 336 µmol/L (3.8 mg/dL) in the control group. The composite endpoint of kidney failure or death was reached by 100 (28.4%) versus 109 (31.0%) in the PLEX and control groups, respectively, which indicated a nonsignificant difference. Oral and intravenous cyclophosphamide were adjusted for age and kidney function and were used in 34.2% and 50.3% of patients, while the other 109 patients received rituximab.[23] Subgroup analyses focusing on the number of participants remaining dialysis dependent, and the magnitude of kidney function recovery is ongoing and will hopefully answer key questions pertinent to the effectiveness of PLEX. One of these projects focused on the 61 patients (8.7%) who had severe alveolar hemorrhage at the time of inclusion, which was defined as oxygen saturation ≤85% on room air or requiring mechanical ventilation. Among those, the addition of PLEX nonsignificantly reduced the mortality rate at 3 and 12 months of follow-up (16.1% and 19.4% vs. 30.0% and 33.3%).[128a]

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 32.7 Summary of Studies Reporting on Combined use of Cyclophosphamide and Rituximab

Trial/Study	RITUXVAS[122]	Pepper et al.,[125]	Gulati et al.,[124]	Cortazar et al.,[123]
Patients	33 and 11 (RCT)	49	64	129
Induction	PLEX (8 and 3 patients), *RTX arm* (4-times 375 mg/m^2 and 2 pulses CYC, 15 mg/kg with adjustment), *CYC arm* (according to CYCLOPS[a]), followed by azathioprine maintenance	RTX (2-times 1000 mg) CYC (6 pulses IV, 500 mg each or 500-750 mg)	PLEX (7 sessions), RTX (2-times 1000 mg) CYC (IV, 10 mg/kg, max. 750 mg first 2 pulses, 4-times 500 mg subsequent)	PLEX (40 patients) RTX (2-times 1000 mg) CYC (oral, adjusted)
Glucocorticoids	1 g methylprednisolone (+ optional doses), oral prednisolone down to 5 mg at month 6	1-2 pulses methylprednisolone 250 mg-1 g, oral prednisolone for 1-2 weeks	2.6 g (6 months), initial dosage max. 60 mg/day	Pulse methylprednisolone (optional), oral prednisone taper over 7 months
Maintenance	CYC arm only (AZA)	AZA (first line), MMF, MTX, RTX	MMF or AZA	RTX (1000 mg every 4 months)
Serum creatinine (µmol/L)[b]	—	179 and 174 (2 groups)	558	—
eGFR (mL/min/1.73 m^2)	20 and 12	28.9 and 28.3 (2 groups)	9	18.8 (75 patients with "RPGN")
Remission (at 6 months)	76% and 82% (sustained remission)	96% (47 patients)	94% (60 patients)	84% (109 patients)
Kidney recovery	eGFR 39 and 27 mL/min/1.73 m^2 at month 12	Serum creatinine 109 µmol/L and 113 µmol/L at month 12	67% (of 30 with KRT at baseline); of non-KRT patients, eGFR improved to 45 mL/min/1.73 m^2 at 3 years	eGFR 28.9 mL/min/1.73 m^2 at month 6 (PR3 significantly better)
Mortality	6 and 2 (18% each)	3 (6.1%) of which 1 was an infectious complication	9 (14.1%), of which 5 were infectious complications (1 COVID-19)	4 (3.1%)

[a]CYCLOPS[126]: 15 mg/kg body weight cyclophosphamide biweekly for the first 3 infusions, then 3 further infusions 3 weeks apart; potentially extension to 10 infusions in case of insufficient response. Maximum dose per pulse 1.2 g; adjust for age: 60-70 years (–2.5 mg/kg), >70 years (–5 mg/kg); adjust for kidney function: 300-500 µmol/l (3.4-5.7 mg/dL) – 2.5 mg/kg; and adjust for potential leukocyte nadir: leukocyte nadir 2-3 x 10^9/L – 20% and nadir 1-2 x 10^9/L – 40%.

[b]To convert the values for creatinine to milligrams per deciliter (mg/dL), divide by 88.4.

The RITUXVAS trial was a randomized controlled trial (3:1 randomization), while the others were retrospective observational studies with the limitations pertinent to this study design.

AZA, Azathioprine; *COVID,* coronavirus; *CYC*, cyclophosphamide; *eGFR*, estimated glomerular filtration rate; *IV,* intravenous; *MMF,* mycophenolate mofetil; *MTX,* methotrexate; *KRT,* kidney replacement therapy; *PLEX*, plasma exchange; *PR3*, proteinase 3; *RCT,* randomized controlled trial; *RPGN*, rapidly progressive glomerulonephritis; *RTX*, rituximab.

A subsequent meta-analysis incorporating both key trials found that among 1060 participants, there was no effect on all-cause mortality when PLEX was added. Analysis of 7 trials including 999 patients (84.5% from MEPEX and PEXIVAS) found that PLEX reduced the risk of kidney failure at 12 months by 38%, while data on 908 participants revealed that it increased the risk of serious infections within 1 year by 27%.[129] The number needed to treat (NNT) when a patient presents with a serum creatinine >500 µmol/L (5.7 mg/dL) to prevent a case of kidney failure at 1 year was 6.9, while the number needed to harm (NNH, serious infection) was 7.4. The meta-analysis indicated that a NNT of 23.8 is required to prevent one kidney failure case when serum creatinine is >300 µmol/L (3.4 mg/dL), while the NNH only marginally increases to 11.8.[129] An ad-hoc recommendation stated a weak recommendation to use PLEX when the serum creatinine is >300 µmol/L, as these patients were considered at moderate to high risk to develop kidney failure.[130] The decision whether to perform PLEX will remain an individual center decision and will be influenced by the experience of the center with extracorporeal therapies as it is likely that expert centers are able to better mitigate the risks associated with PLEX.

In the PEXIVAS trial, 191 patients (27.1%) also had pulmonary hemorrhage at baseline, of whom 61 (31.9%) were considered severe, which was defined as oxygen saturation of ≤85% (breathing ambient air) or requiring the use of mechanical ventilation.[23] In the 130 patients with nonsevere diffuse alveolar hemorrhage, 2 patients in the PLEX and 5 patients in the non-PLEX arm died (HR 0.43, 95% CI 0.08–2.31), while in those 61 patients with severe diffuse alveolar hemorrhage, mortality rates were 19.4% (*n*

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

= 6) and 33.3% (n = 10) in the respective treatment arms (HR 0.45, 95% CI 0.14–1.40). The large confidence intervals underline the lack of significance.

Some centers still perform PLEX in those with severe alveolar hemorrhage. Also, in patients with the most severe kidney disease (e.g., presenting with a serum creatinine ≥3.4 mg/dL and signs of active ANCA-GN), it might be considered as also mentioned by the most recent EULAR and KDIGO recommendations/guidelines.[119,131]

MAINTENANCE OF REMISSION

Remission is defined by BVAS of 0 (i.e., absence of active disease manifestations). Therapies used to induce remission, especially cyclophosphamide, should not be continued during the remission stage due to toxicity of long-term use. The ultimate goal is to avoid disease relapses, and factors associated with disease recurrence (PR3-ANCA–positivity, normal kidney function, ENT disease, prior relapse, short-term therapy, and a transition from negative-to-positive ANCA after negativity) need to be incorporated into individualized treatment protocols, especially the length and duration of maintenance of remission. As a first step in the modern era of maintenance therapy, the CYCAZAREM trial revealed that the substitution of cyclophosphamide by azathioprine was safe and effective, as 11 relapses occurred in the azathioprine (15.5%) and 10 in the cyclophosphamide group (13.7%; P = 0.65).[132] Further studies revealed that methotrexate was noninferior to azathioprine in maintaining remission (33.3% vs. 36.5%, P = 0.71), while patients receiving mycophenolate mofetil for maintenance of remission had a higher relapse rate than those treated with azathioprine (55.3% vs. 37.5%, P = 0.03).[133] Methotrexate, however, has a narrow therapeutic range in patients with ANCA-GN and is thus not a valuable option for many of those with kidney disease. These therapies have been replaced by rituximab and specific trials are discussed later.

The MAINRITSAN program conducted by the French Vasculitis Study Group and the RITAZAREM trial by an international consortium consolidated rituximab as the first-line therapy for the maintenance of remission. MAINRITSAN1 randomized patients in remission after cyclophosphamide induction to either rituximab, given at a dose of 500 mg twice at trial inclusion, and then one infusion every 6 months for three further infusions, or azathioprine maintenance therapy (2 mg/kg for 12 months, then 1.5 mg/kg for 6 months, and then 1 mg/kg for another 4 months). MAINRITSAN2 tested a tailored approach of rituximab administration on the basis of recurrence of CD19-positive B cells in peripheral blood or a change in ANCA status, from negative to positive or a twofold increase compared with the five-dose regimen used in MAINRITSAN1. This idea derived from observational data showing that B cell repopulation is associated with disease flare[134] and that a rise in ANCA titers can influence the risk of disease relapses, especially kidney relapses.[135] MAINRITSAN3 rerandomized patients from MAINRITSAN2 after 28 months to receive either further 6 monthly rituximab infusions or placebo for 18 months. In brief, in MAINRITSAN1, rituximab was superior to azathioprine for prevention of major relapses (5% vs. 29%; P = 0.002) and had a comparable safety profile.[136] In MAINRITSAN2, the tailored approach allowed for reduction of rituximab infusions from 5 to 3, but relapses were more common, albeit nonsignificantly, in the tailored rituximab arm (17.3% tailored; on average 3 infusions; vs. 9.9% 5-dose regimen; P= 0.22).[137] All major recommendations/guidelines advise against the use of a tailored approach, as there is a clinically meaningful difference in relapse rate, plus the tailored approach seems impractical to implement in clinical routine. MAINRITSAN3 included 50 and 47 patients in remission after completion of MAINRITSAN2 and randomly assigned these patients to receive either further rituximab (4 infusions, every 6 months) or placebo. The patients were followed for 28 months and relapse-free survival was 96% versus 74% with rituximab versus placebo (P = 0.008). The rate of serious adverse events was nonsignificantly higher in the rituximab arm.[138] In the RITAZAREM trial, only patients with relapsing disease were recruited (n = 188) and received rituximab and glucocorticoids as induction therapy. Of the 188 patients, 170 patients achieved remission and were then randomized to rituximab or azathioprine as maintenance therapy. These patients in the rituximab arm received a total of 5 doses 4 months apart and at a high dose (1000 mg each) or azathioprine at 2 mg/kg/day, tapered after month 24. An excess in relapses (32 vs. 13) was observed in the azathioprine group (P < 0.001). The rate of serious adverse events was, in line with the MAINRITSAN results, comparable, although more patients receiving rituximab developed moderately severe acquired hypogammaglobulinemia, which was defined as an IgG level <5 g/L.[139] A comparison of a strategy to administer rituximab based on either B cell repopulation or a significant rise in the ANCA titer (MAINTANCAVAS trial) found that only 4.1% of patients who were randomized into the B cell arm relapsed in comparison with 20.5% in the ANCA arm over an observational period of 4.1 years. Noteworthy, a mean number of 3.6 infusions were required in the B cell arm in comparison with only 0.5 infusions in the ANCA arm.[141] This at first sight contrasts with the findings of MAINRITSAN2,[137] but in MAINRITSAN2 the experimental arm received 3 infusions over 18 months, thus likely having a profound B cell depletion over the observational period. In addition, the threshold of ANCA rise has been different between the trials (e.g., a 5-fold and 4-fold rise and at least 4-times and 2-times above the cutoff value of the assay for MPO- and PR3-ANCA in MAINTANCAVAS[141] vs. doubling in ANCA titers or negative-to-positive transition of ANCA, among others, in MAINRITSAN2[137]). Thus it seems obvious that patients in MAINTANCAVAS were undertreated, provoking disease relapses. All recommendations/guidelines suggest using fixed regimens rather than tailored ones.

Duration of Maintenance Therapy

The updated EULAR recommendations state that maintenance therapy should be continued for 24 to 48 months, which also includes the induction phase of therapy.[119] However, KDIGO 2024 mentioned that the certainty of evidence for continuing rituximab beyond 18 months of maintenance was graded as very low due to serious imprecision.[131] The recommendations to use a longer course of rituximab are largely based on the results of MAINRITSAN3, providing evidence that continuous administration of rituximab reduces relapses,[138] and the REMAIN trial, which randomized patients to either continuous glucocorticoids (withdrawal after 23 months) and azathioprine (until end of follow-up) or withdrawal of

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

glucocorticoids (after 4 months) and azathioprine (after 2 months).[142] Continuation of azathioprine was associated with a lower relapse rate (22% vs. 63%; $P < 0.0001$), and patients had a better kidney prognosis, but the patients receiving continuous azathioprine had, in turn, more serious adverse events.

Maintenance therapy might be offered for an even longer period of time to certain patient groups (i.e., those with frequent relapses in the past and especially those wishing to receive continuous therapy, as a part of shared decision making). This includes an individualized discussion with the patient about risk of relapse and the risks attributable to disease relapses (e.g., development of CKD and risk of kidney failure). These considerations need to be balanced against the potential risks associated with long-term use of immunosuppression. Prolonged rituximab administrations frequently lead to hypogammaglobulinemia, which is observed in up to 50% of patients with AAV.[143] A more tailored approach to continue immunosuppression may be based on ANCA serology. Patients with MPO-ANCA vasculitis have a lower risk of relapse during follow-up.[22] A retrospective study including 159 patients with MPO-ANCA vasculitis investigated the role of immunologic remission after induction therapy with cyclophosphamide, rituximab, or mycophenolate mofetil. A persistence of a negative MPO-ANCA test was associated with the absence of disease relapses during follow-up, while reappearance of ANCA-positivity was the strongest independent risk factor for relapses.[144] Achieving ANCA negativity within the first 180 days of treatment was associated with fewer relapses during follow-up in a large single-center study, while it had no impact on a composite of kidney failure or death.[145] These data, in general, argue that in certain circumstances therapies to maintain remission might be stopped earlier, but this needs to be in accordance with patient preference and patients require close follow-up. There is a high risk of disease recurrence once immunosuppression is stopped, and this needs to be discussed with the patient on an individualized basis. Those with new disease might never relapse after maintenance therapy is stopped, but those with a relapsing disease course will likely continue this way after withdrawal of immunosuppression. In certain scenarios, treatment might be stopped even earlier than after 24 months. Patients with MPO-ANCA negativity have a minimal relapse risk,[144] along with those established on dialysis. The latter patient population is also at risk of developing severe infections[146]; thus these patients should be managed on the basis of extrarenal disease manifestations, and renal-limited forms will not require maintenance therapy.

KIDNEY PROGNOSIS IN VASCULITIS

The number of patients with ANCA-GN having kidney failure remains substantial. In a series of 212 patients with ANCA-GN, 48 (22.6%) required kidney replacement therapy (KRT) at baseline, of whom 23 (47.9%) regained dialysis-independent kidney function.[147] Such an outcome has relevance because dialysis independence improves overall prognosis, quality of life, and other health care–related outcomes. One of the issues in the management of ANCA-GN is the ill-defined term “kidney function recovery,” but this would certainly include an increase in eGFR from the time of treatment initiation. The potential to recover kidney function depends on baseline eGFR, acuity of presentation, age at presentation, and if patients had established CKD (either due to previous episodes of active disease or other causes). In the RAVE trial, 102 out of 197 patients (52 %) had kidney disease at inclusion. The eGFR of the two treatment groups differed at baseline, with a mean eGFR of 41 mL/min/1.73 m^2 in the rituximab and 50 mL/min/1.73 m^2 in the cyclophosphamide/azathioprine group at baseline. Both groups showed a comparable eGFR increase at 18 months of follow-up, of 8 and 7 mL/min/1.73 m^2,[148] respectively. In the ADVOCATE trial, as outlined earlier, there was a clinically meaningful difference in eGFR increase by 7.3 mL/min/1.73 m^2 (from 44.6) in the avacopan group as opposed to 4.1 mL/min/1.73 m^2 (from 45.6) in the prednisone group at 12 months. This difference became more pronounced when focusing on 27 and 23 patients presenting with an eGFR of below 20 mL/min/1.73 m^2 at baseline, as avacopan led to an increase of 16.1 mL/min/1.73 m^2 in comparison with 7.7 mL/min/1.73 m^2 in the control arm.[116] We argue that the future of clinical trials will require refinement of definitions of kidney outcomes, possibly moving away from hard endpoints such as kidney failure or death as a combined endpoint. A greater increase in eGFR might in turn also improve long-term kidney outcomes. Increase in eGFR might be a more powerful and meaningful endpoint in clinical trials than assessment of renal BVAS items during subsequent follow-up. Another issue is that no study thus far has focused on the natural course of kidney disease in patients with ANCA-GN. In the UK RaDaR study focusing on rare kidney disease, the eGFR decline in those with ANCA-GN seemed to be paralleling what is expected in the general population, a decline of 1 mL/min/1.73 m^2 in an aging population.[149]

Preventing renal relapses is another important step to reduce the burden of kidney failure in patients with GPA or MPA. A retrospective analysis of 535 patients recruited into earlier trials conducted by the European Vasculitis Society (EUVAS) reported the occurrence of 101 relapses. Apart from a baseline serum creatinine of over 500 μmol/L (5.7 mg/dL; hazard ratio 13.8), the occurrence of renal relapses was the strongest risk factor for development of kidney failure, with a hazard ratio of 8.9.[150] These findings underline the clinical imperative of preventing renal relapses. An analysis of 571 patients (77% had kidney disease) recruited into RCTs conducted by EUVAS or the French Vasculitis Study Group found that 34.3% had persistent proteinuria, defined as urine-protein-to-creatinine ratio of ≥0.05 g/mmol, and 29.8% (of all patients) had hematuria at the end of induction therapy. Persistent presence of proteinuria and hematuria predicted renal relapse during follow-up, while only proteinuria was associated with kidney failure during follow-up.[151] Uncertainty exists whether ongoing hematuria also indicates ongoing glomerular inflammation or whether it could be merely a sign of chronic damage. A small single-center study performed repeat kidney biopsies in 57 patients with suspected active disease and found an agreement in only 41% of performed biopsies,[152] indicating that a significant proportion of patients will have persistent hematuria due to chronic glomerular damage rather than active disease. Future research will need to determine whether urinary microscopy, especially hematuria of glomerular origin (e.g.,

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

red blood cell casts), would be a helpful tool to predict kidney disease activity more accurately.

OVERALL PROGNOSIS

In the 1970s, patients with MPA had a 5-year mortality rate of >89% if no treatment was offered to patients.[13] Analysis of data from the Danish registry found a 5-year mortality rate of 18.6% when patients were diagnosed between 2010 and 2015. The leading causes of death were cardiovascular disease (5.3%), infections (4.0%), malignancies (2.6%), and respiratory disease (2.2%).[14] Data from 4 European Vasculitis Study Group (CYCAZAREM, NORAM, CYCLOPS, and MEPEX) clinical trials conducted more than 2 decades ago reported infectious complications as the leading cause of death during the first year of follow-up, while mortality was driven by cardiovascular disease, malignancy, and infections thereafter. Independent risk factors in the analysis of 535 patients were age and eGFR <15 mL/min/1.73 m^2.[153] Assessment of long-term follow-up of the original cohort in combination with the addition of another 313 patients from the RITUXVAS, MYCYC, and IMPROVE trials revealed male sex as another risk factor for mortality. Moreover, it became apparent that severity of kidney dysfunction at initial presentation and treatment burden (i.e., more intense immunosuppression) impacts mortality, as patients in the MEPEX and RITUXVAS trials had the worst prognosis, while those with no or only mild kidney function impairment at baseline, such as patients randomized to participate in the NORAM, IMPROVE or MYCYC trials, had a better overall prognosis.[154]

The Danish national registry data revealed that overall prognosis has improved over the years. A 5-year mortality rate of 27.6% was reported between 2000 and 2004, which fell to 18.6% during the years 2010 to 2015. Moreover, in comparison with a matched background population, AAV patients requiring kidney replacement therapy and those who developed CKD had higher mortality rates (adjusted risk ratios of 3.5 and 2.6, respectively) in comparison with those with no kidney disease at initial discharge (adjusted risk ratio of 2.0).[14] The excess mortality observed in patients with kidney failure is mainly driven by infectious complications and cardiovascular events (61%), while vasculitic flares are an infrequent cause of death (6%).[146] The benefit/risk ratio is skewed toward more complications due to immunosuppression in this vulnerable population and initiation or withdrawal of maintenance immunosuppressive therapy needs to be based on shared decision making,[155] taking into account other organ manifestations besides the kidney. Other factors also associated with poor prognosis include UIP,[156] especially encountered in MPO-ANCA vasculitis, and a low serum C3 level at baseline,[62] which independently predict mortality.

COMPLICATIONS AND RISK MANAGEMENT

Overall, excess mortality remains an issue in AAV, and this especially affects patients with kidney disease. Serious infections remain a concern in the management of AAV. A retrospective study of 192 patients who received rituximab indicated that the addition of trimethoprim-sulfamethoxazole prophylaxis reduces the occurrence of severe infections by 70%.[157] Analysis of the RAVE trial confirmed this finding and found that trimethoprim-sulfamethoxazole, in contrast to other measures for *Pneumocystis jirovecii* pneumonia prophylaxis, reduces the frequency of overall serious infections.[158] While trimethoprim-sulfamethoxazole as a prophylactic measure is standard of care in patients with AAV, other interventions to reduce the burden of infectious complications are ill defined in patients with glomerulonephritis.[118]

CKD is common in ANCA-GN and depends on disease manifestations (rare in those without kidney disease at baseline) and the severity of initial acute kidney disease.[159] Measures used as "nephroprotection" have not been tested systematically in patients with ANCA-GN, as AAV and immunosuppression were common exclusion criteria in trials. There is a strong rationale to use sodium glucose co-transporter 2 (SGLT2) inhibitors in this patient population, given their risk of kidney failure and cardiovascular disease,[160] but trial data are not likely to be generated in the near future and information from real-life observations on SGLT2i safety and efficacy in AAV remains limited.[161]

Cardiovascular events are common in patients with AAV. Traditional and disease-specific cardiovascular risk factors contribute to this increased risk. An analysis of 2286 patients with AAV found that age, current smoking status, and pulmonary and kidney involvement are associated with cardiovascular events.[16] It has been shown that patients with AAV in remission evidence a 30% reduction in endothelium-dependent vasodilation, a 50% reduction in acute release of endothelial active tissue plasminogen activator, an increased arterial stiffness, and a twofold increase in plasma endothelin-1.[162] This combination of findings is referred to as diffuse endothelial dysfunction. As a hallmark of active vasculitis, this persists during phases of disease "remission"[163] and highlights that patients might indeed have ongoing subclinical inflammation, which is not detected by commonly performed serologic and imaging tests.

Lipid levels in patients with a new diagnosis of AAV, especially those with PR3-ANCA vasculitis, are lower at baseline. Lipid levels increase when remission is achieved.[164] Reasons for this finding remain obscure, but a longer time to a final diagnosis with persistence of inflammation and energy consumption and muscle wasting might play a relevant role. The cardiovascular risk profile of patients with AAV should be assessed on an annual basis, although there is no consensus on which investigations should be performed. A survey of patients in Germany revealed that treatment targets to manage hyperlipidemia and hypertension are not met in most patients with a high-risk constellation.[165] The recommended target levels for cholesterol and blood pressure were lowered recently in international recommendations/guidelines and include the fact that patients with CKD are at increased risk of cardiovascular events. The risk in patients with ANCA-GN may be even higher because of disease-specific changes, as outlined earlier, which may potentiate the risk of cardiovascular events above and beyond their respective CKD class. Indeed, patients with AAV have been observed to have a >twofold increased risk of cardiovascular events compared with a matched CKD population,[166] further highlighting the need for more vigilant control of modifiable cardiovascular risk factors in this high-risk population.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

KIDNEY TRANSPLANTATION

A substantial proportion of patients with ANCA-GN will eventually require kidney replacement therapy and should be evaluated for a deceased-donor or living-related kidney transplantation. The most important questions relate to how long patients should wait after achieving remission before they are eligible for a kidney transplant (i.e., time from diagnosis to "quiescent" disease) and whether patients will relapse after receiving a transplant. Analysis of a European dialysis and transplant registry between 1993 and 2012 reported a stable burden of kidney replacement therapy initiation in patients with AAV. Of these, 618/2511 (24.6%) received a first kidney transplant. This low rate of transplantation might be related to the age of patients with ANCA-GN, averaging 67 years.[167] A large multicenter study of 85 patients with AAV who received a kidney transplant showed that most patients were ANCA-negative (66%) at the time of transplantation. The relapse rate was 0.02 per patient-years, and relapses were encountered in 7 out of 85 patients. Although not significant, more patients in the GPA group (5/42) and more patients with ANCA-positivity at the time of transplantation (4/29) had a relapse during follow-up.[168] In another multicenter study with 110 patients, the posttransplantation recurrence rate of ANCA-GN was 2.8% per patient year, accumulating to a total of 11 of 110 patients within the first 5 years after transplantation. Four of these 11 patients lost their graft, with overall 1-year and 5-year graft survival rates of 94.5% and 82.8%, respectively. In multivariate analysis, recurrent disease was independently associated with subsequent graft loss.[169] Kidney transplantation might be considered when patients are in stable remission for a duration of ≥12 months.[170] Notably, the prognosis of patients receiving a kidney transplant overall is favorable, as has been highlighted in the European registry, and it should be offered to all patients deemed to be eligible to receive a graft. The rate of recurrence of disease seems to be low and is especially seen in those with a persistent-positive ANCA test and individuals with a diagnosis of PR3-ANCA vasculitis, mimicking the risks observed in patients with native kidney function. Analysis of USRDS data revealed that the risk of nonfatal/fatal myocardial infarction and stroke was reduced by 78% in patients with AAV receiving a kidney transplant in comparison with those remaining waitlisted, and these effects persisted after adjustment for sex and age groups,[171] further highlighting the impact of kidney transplantation on overall prognosis.

ANTIGLOMERULAR BASEMENT MEMBRANE GLOMERULONPHRITIS

INTRODUCTION

Antiglomerular basement membrane (GBM) disease, historically known by the eponym Goodpasture disease, is a rare but life-threatening autoimmune disorder. In 2012, it was included as an immune-complex small vessel vasculitis in the revised Chapel Hill Consensus Conference (CHCC) nomenclature.[8] The disease is caused by circulating pathogenic autoantibodies directed against components of the glomerular and the alveolar basement membrane. Acute kidney injury is its most common manifestation. Kidney involvement is almost invariably present while lung involvement occurs in approximately 50% of cases. Presentation is often abrupt and dramatic, while forms with a more benign phenotype do exist, sometimes without detectable circulating antibodies by commercially used tests.[172] Environmental factors, in the background of genetic predisposition, are implicated in the pathogenesis. Linear deposition of antibodies along the GBM with concomitant diffuse crescent formation are the histologic hallmarks. It is a one-hit disease with a low recurrence rate.[173]

EPIDEMIOLOGY

Anti-GBM disease occurs across all ethnicities, albeit with variation. An observational study from Ireland reported 79 cases between 2003 and 2014, amounting to a national incidence of 1.6 per million per year.[174] The disease affects all age groups but is uncommon in children.[175] Age distribution is bimodal with a first peak in adolescence and early adulthood and a second peak later in life, usually in the seventh decade. Younger patients frequently show a pulmonary-renal presentation, while the disease often exhibits a renal-limited course in older individuals.[176] There is a slight preponderance among males in younger persons, while there appears a small female predominance in the older age group.[177] There is a general impression that the incidence may be increasing over the past decades.[178]

Consistent with a genetic predisposition, there is strong HLA-linkage with HLA MHC class II *DRB1*15:01* in about 80% of cases, whereas *DRB1*01* and *DRB1*07* are protective.[179] Given a high background prevalence of the former allele in the general population of around 10%, additional factors must have a role in disease initiation. Despite these genetic associations, familial clustering is extremely rare. Recently, Durcan and colleagues reported two siblings, aged 60 and 71 at the time of diagnosis. Both individuals were heterozygous for HLA *DRB1*15:01* and *DRB1*04:03*.[179a] Another study of familial anti-GBM disease found a genetic variant of type IV collagen as a potential structural risk factor for anti-GBM disease.[180]

Anti-GBM disease can been triggered by certain drugs and industrial solvents. Alemtuzumab, a monoclonal antibody directed against CD52, used for relapsing multiple sclerosis, has been implicated as an agent causing anti-GBM disease in a small subset of exposed patients.[181] More recently, immune checkpoint inhibitors have been implicated as precipitants of anti-GBM disease, predominantly in its renal-limited form. A summary of published cases reported positive autoantibodies in five of seven cases and renal outcomes were consistently poor.[182] There is an association of infections with the onset of anti-GBM disease, and about half of the patients experience prodromal infections.[183] Temporal clustering has also been reported in several series during the COVID-19 pandemic.[184] Likewise, there have been reports of anti-GBM disease manifesting in temporal association with various vaccines, such as pneumococcal or influenza, or,

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

more recently, after vaccination against SARS-CoV2.[185] Anti-GBM disease has been observed after certain urologic procedures such as lithotripsy.[186] Mechanical exposure of previously hidden antigens may explain this phenomenon.

PATHOLOGY

Kidney biopsy establishes the diagnosis and also provides valuable prognostic information. There is general consensus that a biopsy should be performed whenever possible.[187] Strong linear immunoglobulin deposition along the GBM can be visualized by direct immunofluorescence (Fig. 32.6) and is considered the pathologic hallmark of the disease, almost invariably associated with cellular crescents (often exceeding 80%) and fibrinoid necrosis on light microscopy. In contrast to biopsies from AAV patients, lesions observed on histopathologic examination are usually of the same age (see Fig. 32.6). The immunoglobulin is usually of the IgG type (mostly IgG1 subclass), but isolated cases of an IgA-dominant pattern have been reported.[188] Additional complement deposition (C3) is usually present. Nasr and colleagues[189] reported two cases of IgM anti-GBM nephritis without C3 deposition out of a total of 20 distinct patients with a comparably benign course. No serum autoantibodies were detected, and no pulmonary manifestation occurred; other characteristics of this cohort are discussed later. A report by Antonelou and colleagues[190] described two additional cases of anti-GBM disease mediated by IgM antibodies, who had a more severe clinical and histologic phenotype.[190] Overall, non-IgG anti-GBM disease is exceedingly rare and may pose diagnostic challenges.

PATHOGENESIS

Anti-GBM disease represents a prime example of an immune-mediated condition. Compelling evidence for underlying autoimmunity stems from animal models performed in the 1960s. In a classic transfer experiment, Lerner and colleagues[191] injected autoantibodies eluted from kidneys of patients into monkeys, thereby triggering glomerulonephritis. Some 20 years later, Wieslander and colleagues[192] reported that the target antigen ("Goodpasture autoantigen") is the noncollagenous domain 1 (NC1) of the α3 chain of type IV collagen (α3[IV]NC1) in the basement membrane. Six distinct α-chains (α1-6) constitute the collagen IV family; they form three triple-helices, which polymerize to form the lattice-like structure of the basement membrane.

The expression of one such protomer (α3 chain) is essentially confined to the glomerular and alveolar basement membrane, in keeping with the clinical spectrum of the disease. An intriguing observation was the discovery of naturally occurring low-level autoantibodies to α3[IV]NC1 in healthy individuals.[193] In a binational cohort study, autoantibodies with low affinity were detected in the serum

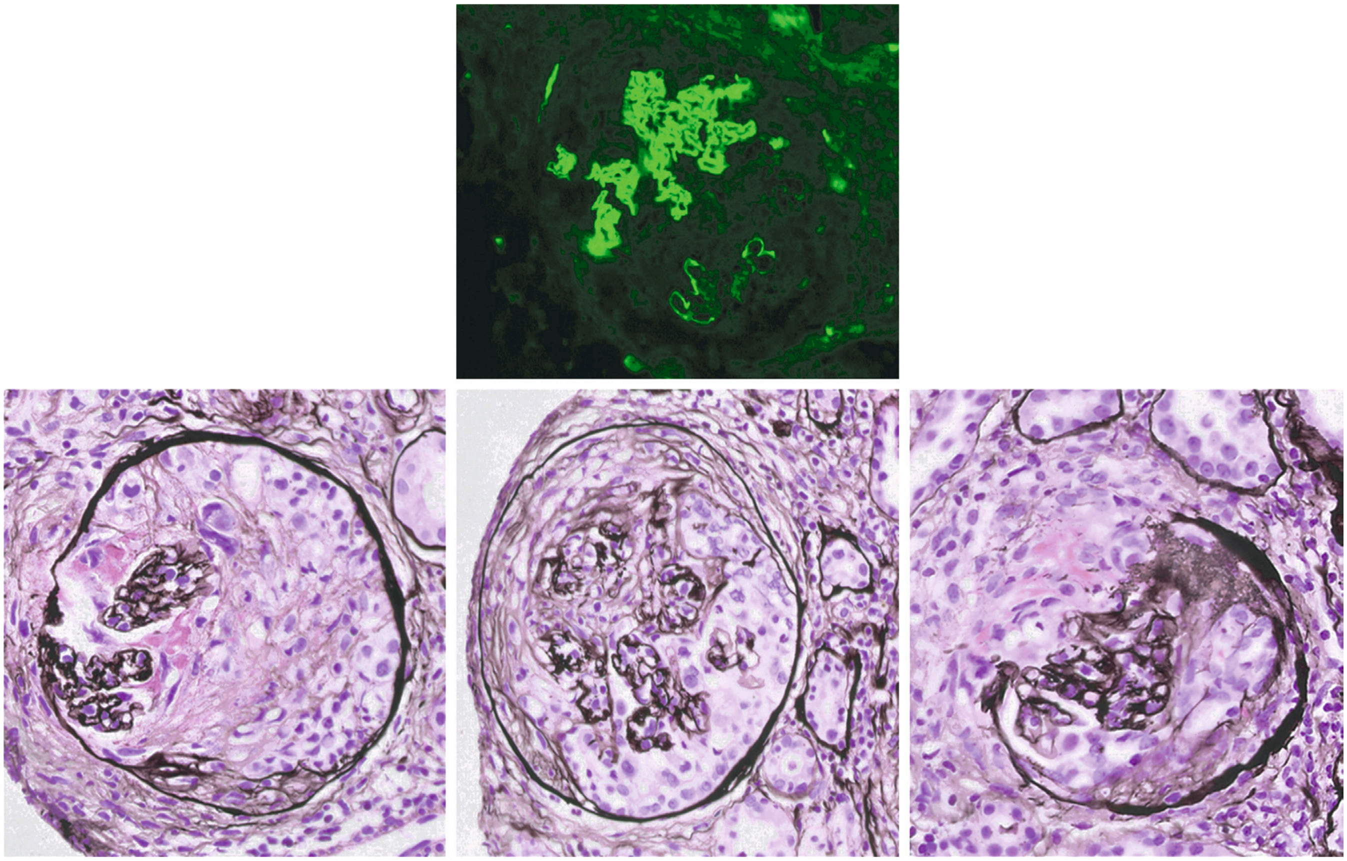

Fig. 32.6 Antiglomerular basement membrane disease: typical linear staining for IgG (top) in combination with crescentic glomerulonephritis (GN). Glomeruli are usually affected in the same stage of crescentic GN. Crescents may be cellular, fibrocellular, or fibrous (lower panel, from left to right).

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

of 20 blood donors. Epitope specificity was identical to those in patients, but IgG subclasses differed. While IgG1 and IgG3 predominated in affected individuals, subclass restriction to IgG2 and IgG4 was noted in the healthy probands. Anti-GBM antibodies may precede the onset of disease. A case control study by Olson and colleagues[194] further helped to better delineate the natural course of the disease. Using serum samples from the Department of Defense Serum Repository (30 patients diagnosed with anti-GBM disease and 30 matched controls), the authors found subclinical autoantibody production months or even years before the onset of disease. Despite this major role of humoral immune response, an important contribution of autoreactive T cells in initiation and progression of anti-GBM disease is evident, and T cell infiltrates (mainly CD4) are present in kidney biopsies of patients with active disease.[195] Multiple steps are implicated in the eventual loss of immunologic self-tolerance. Apart from immunologic factors outlined earlier, one key component in pathogenesis is a structural change of the α3[IV]NC1 antigen. Various pathogenic events may disrupt the structure of the GBM, causing exposure of previously cryptic antigenic epitopes (denoted E_A and E_B), which subsequently interact with preexisting circulating autoantibodies. Subsequently, anti-GBM antibodies deposit along the GBM, resulting in an inflammatory process and diffuse crescent formation. Anti-GBM disease is therefore considered an autoimmune conformeropathy.[196]

ANCAs are present in the serum of about one-third of patients with anti-GBM disease while approximately 6% of patients with AAV also have anti-GBM antibodies.[61] Such "double-positive" serology confers a distinct phenotype, with the acute presentation resembling classical anti-GBM disease, while the long-term course shows a propensity for relapses, more in keeping with isolated AAV. A large multicenter European study reported a substantially higher prevalence of anti-MPO antibodies than anti-PR3 antibodies in such individuals (70% vs. 27%).[61] While the coexistence of ANCA was previously considered a favorable prognostic marker in anti-GBM disease,[197] a large cohort study of 174 patients (38.5% double positive) found that the amount of normal glomeruli, rather than serotype per se accounted for the better outcome associated with double positivity.[198]

Novel target antigens have been reported in association with anti-GBM disease.[195] Autoantibodies against peroxidasin, a heme peroxidase expressed within the GBM, exhibiting significant homology to MPO were found in a subset of patients with anti-GBM disease and might have a contributory role in pathogenesis.[199] Shen and colleagues[200] proposed Laminin-521 (LM521), another key component of the mature GBM, as a new autoantigen in anti-GBM disease. In a retrospective study of 101 patients with anti-GBM disease, the authors discovered autoantibodies against LM521 in about a third of patients, while none were present in a matched control group. ANCA-positivity (predominantly MPO-subtype) was found in about 30% of LM521-negative patients but only in 12% of patients who tested positive for LM521. Despite the higher age (median 54.0 years), pulmonary hemorrhage was present in more than half of all LM521-positive cases. A follow-up study confirmed the nephritogenicity of these autoantibodies in both rodents and humans.[201] These reports indicate that a subset of patients may present with additional antibodies associated with a more severe phenotype, with implications for risk stratification at baseline.[202]

CLINICAL FEATURES AND NATURAL HISTORY

Many patients will report nonspecific prodromal symptoms before a diagnosis of anti-GBM disease is made. These include general malaise, fever, weight loss, and arthralgia.[203] When compared with other forms of vasculitis, such as AAV, the duration of prodromal stage is frequently shorter (weeks instead of months).[61] Infections are implicated as triggers of outbreaks and frequently coexist at presentation.[183]

Two broad modes of manifestation can be distinguished: The classical pulmonary-renal syndrome, which dominated early case series, is associated with active smoking status and occurs more frequently in younger individuals. Isolated kidney disease, often rapidly progressing to advanced kidney failure and oligoanuria, is the typical presentation in later life. Patients may report loin pain or frank hematuria.[203] An active urinary sediment is present and proteinuria is common, albeit not usually in the nephrotic range. In most cases, kidney involvement is abrupt and severe, and more than half of the patients are dialysis dependent at presentation.[204] A retrospective cohort study from the United Kingdom of 47 patients found established oliguria (defined as <500 mL/day) as a major predictor of mortality and of poor kidney outcomes after 1 year.[205]

Lung involvement develops in about 50% of affected patients and typically manifests with cough, shortness of breath, and hemoptysis. It occurs predominantly in smokers or individuals with exposure to other pulmonary toxins, such as hydrocarbons.[206] Isolated pulmonary hemorrhage is rare; however, anti-GBM disease should always be included in the differential diagnosis of diffuse alveolar hemorrhage (DAH). Recurrent lung bleeding may also result in substantial anemia. Isolated reports about indolent courses exist in the literature.[207] Anti-GBM disease accounts for 4% of all cases of DAH treated in the intensive care unit, while dual-positive serology is found in another 14%.[208] Seen from the other side, only a minority of patients with pulmonary-renal syndrome have underlying anti-GBM disease because AAV incidence and prevalence rates are higher than anti-GBM disease rates. Importantly, seronegativity does not exclude anti-GBM disease, as there are reports of seronegative disease with predominant pulmonary involvement.[209]

Over the past decade, reports of anti-GBM disease without detectable circulating antibodies by commercial tests have emerged and the disease spectrum now appears wider than previously thought.[172] It is estimated that seronegative anti-GBM disease accounts for approximately 10% of anti-GBM disease.[172,189] Therefore as with seronegative AAV, seronegative anti-GBM disease is a challenging diagnosis. This underscores the importance of defining kidney pathology in combined RPGN and DAH clinical settings. While histopathology exhibits linear Ig deposits as in typical anti-GBM disease, appearance on light microscopy is more diverse. In a series of 20 seronegative patients, diffuse necrotizing glomerulonephritis was absent in all, and only 8 individuals exhibited focal crescents. In this cohort,

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

histologic markers of chronicity were prominent. Kidney function was mildly impaired at presentation, with a median serum creatinine of 1.9 mg/dL (range; 0.9–4.6), and renal and patient outcomes were good.[189]

In majority of cases, anti-GBM disease is monophasic, although antibody rebound after PLEX can be seen. Production of autoantibodies ceases spontaneously after disease manifestation, as illustrated by a cohort of eight patients with kidney-limited involvement, who did not receive therapy; however, this process may take up to 1 year.[210] Recurrence is rare and mainly associated with ongoing tobacco exposure. Relapses following infections, including SARS-CoV-2, have been reported.[211]

LABORATORY FINDINGS

Circulating autoantibodies, one hallmark of anti-GBM disease, can be detected in about 90% of cases by commercial assays (mostly enzyme-linked immunosorbent assays (ELISA) or chemiluminescence assays). Wide implementation of liberal screening (in conjunction with ANCA) in appropriate clinical scenarios greatly facilitates early diagnosis of anti-GBM disease. Conversely, seronegativity (i.e., in the absence of circulating autoantibodies) occurs in up to 10% of cases.[212] Methodologic issues may account for a proportion of these cases. Standard immunoassays are based on the specific epitope NC1 domain of the α3 chain, but some patients may react to different epitopes. Routine assays are usually customized to detect IgG (particularly IgG1) and may therefore miss cases of anti-GBM disease caused by other IgG isoforms, especially IgG_4, and other immunoglobulin subclasses. Of note, patients with a predominance of IgG_4 subclass may represent a unique subset, characterized by female sex and severe pulmonary involvement.[213]

Anti-GBM disease can copresent with other primary glomerulopathies such as IgAN or membranous nephropathy. A series from China identified 15 patients with anti-GBM nephritis with coexisting IgAN, who generally had a better renal prognosis when compared to the “pure” control group.[214] On rare occasions, anti-GBM disease can be found in combination with a histologic pattern of membranous nephropathy. As might be expected, such patients usually exhibit a higher grade of proteinuria (well in the nephrotic range). Non–lupus membranous nephropathy concurrent with anti-GBM nephritis was not associated with any of the known antigens commonly associated with primary membranous nephropathy.[215]

TREATMENT

Left untreated, the prognosis of patients with anti-GBM disease is poor, as exemplified by early case series.[216] In the original publication by Stanton and Tange,[216] most patients died within a few days (mostly of DAH), prompting the authors to refer to this entity as “killing disease of young adults.” Following a landmark publication in the 1970s by Lockwood and colleagues, an observational study involving seven patients that reported the benefit of PLEX when added to immunosuppressive therapy, today’s treatment standard was established.[216a] This approach combines removal of circulating autoantibodies with the suppression of further production of pathogenic antibodies.

Whenever anti-GBM disease is suspected, urgent treatment is required. Corticosteroids remain a central component in the initial management and are usually administered first, pending exact diagnostic assignment by histology and/or autoimmune parameters. Although no evidence exists for the superiority of intravenous high-dose methylprednisolone over oral corticosteroids, this strategy is frequently deployed initially in severe cases. In such circumstances, intravenous pulse methylprednisolone (at a dose of 500 mg per day) is usually administered for 3 consecutive days, followed by oral glucocorticoid therapy (1 mg/kg per day, maximum 80 mg, with subsequent tapering). In a retrospective French study, 81 of 119 patients received pulse intravenous steroids.[217] Likewise, in one trial, pulse glucocorticoids were administered to 14 of the 15 patients included (mostly at a dose of 500 mg per day).[218] However, experienced centers recommend a more restrained approach and advocate oral therapy from the outset.[212] In line with current protocols for AAV, the dose of prednisolone should be below 20 mg per day after 6 weeks.[23]

Anti-GBM disease is considered the prototypic indication for PLEX and carries an American Society for Apheresis (ASFA) class I recommendation in the context of DAH and/or dialysis independence.[219] One session of PLEX reduces antibody levels by about 60%. A small clinical trial of 17 patients, where 8 were randomized to receive PLEX (4-L plasma volume [PV] exchanged every 3 days) on top of glucocorticoids and cyclophosphamide, reported that this procedure was more effective in terms of disappearance of autoantibodies and associated with better renal outcomes in comparison with the group receiving glucocorticoids and cyclophosphamide.[220] This experience aligns with a much larger retrospective study from China ($n = 221$).[221] PLEX (4-L PV exchanges) should initially be performed daily. Albumin can be used as replacement fluid, but fresh-frozen plasma (FFP) should be supplied in cases of DAH, following kidney biopsy, or when fibrinogen levels fall precipitously. High-frequency PLEX carries the risk of dilutional coagulopathy, and FFP should be used intermittently, even in the absence of pulmonary bleeding. PLEX is usually continued for 2 to 3 weeks. Autoantibodies should be monitored closely, and a major therapeutic aim is to reduce levels rapidly. PLEX can be discontinued when antibodies are within normal limits or in a “nontoxic range,” which remains ill defined. A nationwide French study reported a median number of 13 sessions (interquartile range [IQR], 9–17).[222] Likewise, an analysis of registry data from the World Apheresis Society reported a median number of 12 sessions with substantial variation, about twice as many as patients with AAV in the same cohort.[223] Cessation of PLEX requires thorough monitoring, as serologic rebound is common and necessitates reinstatement of PLEX. Immunoadsorption is an alternative extracorporeal treatment option and has a higher capacity to remove autoantibodies. A retrospective Austrian study of 10 patients reported antibody negativity after two to nine sessions.[224] To date, experience with this form of therapy remains limited but could be considered, when available, in patients with high autoantibody levels or for individuals with above-average weight.[202]

Cyclophosphamide represents the third pillar of therapy. Treatment can be started orally (2–3 mg/kg per day, adjusted for age, leukocyte count, and kidney function) and

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

is usually continued for 3 months. A general shift toward intravenous cyclophosphamide, in line with management of AAV (according to the CYCLOPS protocol), has been observed. In a retrospective study from the French Society of Hemapheresis of 122 patients diagnosed between 1983 and 2006, two-thirds received intravenous cyclophosphamide.[222] Notably, outcomes were suggestive of superiority of oral dosing, although this might be due to confounding by indication. Given the short duration of exposure, concerns about fertility or increased risk of malignancy are usually not warranted, particularly as the rate of recurrence is low.

Unlike in other autoimmune kidney diseases, the role of anti-CD20 therapy (e.g., rituximab) is not well established. Evidence is limited to case reports and small case series,[225] and the 2021 KDIGO guidelines recommend its use only in refractory cases.[226] The first report about the successful use of rituximab was for relapsing disease.[227] Subsequently, rituximab has been utilized in resistant disease.[228] In this context, it may shorten the time to seronegativity, potentially enabling earlier live-donor kidney transplantation.[212] In 2019, Jain and colleagues[228] provided a review of all published cases treated with rituximab to this date. Out of 22 cases, rituximab was added following conventional therapy for refractory disease in 15, while 7 patients received rituximab initially in lieu of cyclophosphamide. Rapid and complete clearance of autoantibodies occurred in six of them, while the seventh case demonstrated a substantial drop of antibody levels. As the monoclonal antibody is removed by extracorporeal measures, it should be administered after a PLEX session, ideally in a window of at least 48 hours. Such logistic considerations favor a regimen of 2 × 1 g, given 2 weeks apart, if rituximab is considered early in the course of disease. Mycophenolate mofetil might be considered if antibodies remain positive for patients who develop dialysis-dependent kidney failure to facilitate kidney transplantation when patients become seronegative.

A promising addition to the established armamentarium has emerged. Imlifidase is a cysteine endopeptidase derived from the immunoglobulin G (IgG)-degrading enzyme of *Streptococcus pyogenes*. It selectively cleaves the heavy chains of all human IgG subclasses, both circulating and tissue bound, without affecting other immunoglobulins, thereby eliminating Fc-dependent effector functions such as complement-dependent cytotoxicity (CDC) and antibody-dependent cell-mediated cytotoxicity (ADCC). Imlifidase was initially developed as a treatment for desensitizing highly allosensitized adults requiring kidney transplantation, such as those with calculated panel reactive antibodies (PRA) >95%. This action reduces donor-specific antibodies, facilitating transplantation. Following a proof-of-concept study published in 2019, reporting the rapid disappearance of autoantibodies in 3 patients with refractory anti-GBM disease, the GOOD-IdeS-01 trial recruited 15 patients from 5 European countries with anti-GBM disease and severe kidney impairment (eGFR <15 mL/min per 1.73 m^2; with 10 patients requiring dialysis at inclusion). In this open-label pilot trial, a single dose of imlifidase (0.25 mg/kg), given on top of standard therapy, resulted in seronegativity within a few hours. Autoantibody rebound necessitated reinstitution of PLEX in 10 patients because redosing of the enzyme is precluded by the rapid development of anti-imlifidase antibodies.[218] At 6 months, a significantly higher proportion of patients were dialysis independent compared with an earlier study.[61] In line with its mode of action, complete hypogammaglobulinemia occurred temporarily and antibiotic prophylaxis was strongly recommended for the initial treatment period. In all, this publication suggests that imlifidase might assist kidney function recovery, and a phase III trial (GOOD-IdeS-02, NCT05679401) is currently recruiting and close to completion of recruiting the target sample size of 50 patients. The primary endpoint is change in eGFR at 6 months after randomization.

Uncertainty exists regarding the role of immunosuppressive therapy in patients who present with dialysis dependency and/or oligoanuria for ≥48 hours or 100% cellular crescents on a kidney biopsy in the absence of pulmonary involvement.[89,205] In such cases, the prognosis for kidney function recovery is poor and patients are unlikely to derive benefit from intense immunosuppression but face its inherent toxicity. One paper also demonstrated the utility of the renal risk score (RRS; the updated score, AKRiS, is highlighted in Table 32.5),[198] currently used for prognostication in AAV, when applied to patients with anti-GBM disease. Specifically, a percentage of more than 10% of normal glomeruli on biopsy predicted a substantially improved renal prognosis with 33.3% of patients progressing to kidney failure compared with 76.6%, when less than 10% glomeruli were unaffected.[198] Controversy exists whether the initial anti-GBM antibody levels are predictive of renal prognosis, but most studies point toward a correlation between titers and outcome.[229,230] Given the substantial risk of opportunistic infections in patients with glomerular diseases, prophylaxis against *Pneumocystis jirovecii* is recommended.

In classical anti-GBM disease, maintenance therapy is not warranted once remission has been achieved. However, given their propensity for relapse, double-positive cases usually qualify for prolonged therapy, especially in cases of PR3-ANCA-positivity, and azathioprine or rituximab are established treatment options.[202]

Early experience showed a high risk of recurrence if kidney transplantation was performed in the presence of anti-GBM autoantibodies.[231] There is general agreement that transplantation should be deferred until the disease is quiescent and autoantibodies have been negative for 6 months. De novo disease has been observed in patients with X-linked Alport disease after kidney transplantation (Alport posttransplant nephritis). Allosensitization is implicated in the pathogenesis, and the target antigen (collagen IV α5 chain) differs from the classic Goodpasture-antigen; therefore anti-GBM antibodies are often negative by conventional assays. Posttransplant anti-GBM nephritis usually occurs in the first year after transplantation and portends a poor renal prognosis.

Visit Elsevier eBooks+ (eBooks.Health.Elsevier.com) for a complete set of references.

KEY REFERENCES

3. van Daalen EE, Wester Trejo MAC, Göçeroğlu A, et al. Developments in the histopathological classification of ANCA-associated glomerulonephritis. *Clin J Am Soc Nephrol.* 2020;15(8):1103–1111.
5. Rhee RL, Hogan SL, Poulton CJ, et al. Trends in long-term outcomes among patients with antineutrophil cytoplasmic antibody-associated vasculitis with renal disease. *Arthritis Rheumatol.* 2016;68(7):1711–1720.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

10. Kronbichler A, Bajema IM, Bruchfeld A, Mastroianni Kirsztajn G, Stone JH. Diagnosis and management of ANCA-associated vasculitis. *Lancet*. 2024;403(10427):683–698.
12. Gisslander K, White A, Aslett L, et al. Data-driven subclassification of ANCA-associated vasculitis: model-based clustering of a federated international cohort. *Lancet Rheumatol*. 2024;6(11):e762–e770.
17. Wallace ZS, Fu X, Harkness T, Stone JH, Zhang Y, Choi H. All-cause and cause-specific mortality in ANCA-associated vasculitis: overall and according to ANCA type. *Rheumatology*. 2020;59(9):2308–2315.
22. Specks U, Merkel PA, Seo P, et al. Efficacy of remission-induction regimens for ANCA-associated vasculitis. *N Engl J Med*. 2013;369(5):417–427.
23. Walsh M, Merkel PA, Peh CA, et al. Plasma exchange and glucocorticoids in severe ANCA-associated vasculitis. *N Engl J Med*. 2020;382(7):622–631.
32. Watts RA, Hatemi G, Burns JC, Mohammad AJ. Global epidemiology of vasculitis. *Nat Rev Rheumatol*. 2022;18(1):22–34.
36. Olson SW, Owshalimpur D, Yuan CM, et al. Relation between asymptomatic proteinase 3 antibodies and future granulomatosis with polyangiitis. *Clin J Am Soc Nephrol*. 2013;8(8):1312–1318.
43. Xiao H, Dairaghi DJ, Powers JP, et al. C5a receptor (CD88) blockade protects against MPO-ANCA GN. *J Am Soc Nephrol*. 2014;25(2):225–231.
60. Mukhtyar C, Lee R, Brown D, et al. Modification and validation of the Birmingham vasculitis activity score (version 3). *Ann Rheum Dis*. 2009;68(12):1827–1832.
61. McAdoo SP, Tanna A, Hrušková Z, et al. Patients double-seropositive for ANCA and anti-GBM antibodies have varied renal survival, frequency of relapse, and outcomes compared to single-seropositive patients. *Kidney Int*. 2017;92(3):693–702.
75. Hilhorst M, van Paassen P, van Rie H, et al. Complement in ANCA-associated glomerulonephritis. *Nephrol Dial Transplant*. 2017;32(8):1302–1313.
89. Levy JB, Turner AN, Rees AJ, Pusey CD. Long-term outcome of anti-glomerular basement membrane antibody disease treated with plasma exchange and immunosuppression. *Ann Intern Med*. 2001;134(11):1033–1042.
90. Berden AE, Ferrario F, Hagen EC, et al. Histopathologic classification of ANCA-associated glomerulonephritis. *J Am Soc Nephrol*. 2010;21(10):1628–1636.
110. Furuta S, Nakagomi D, Kobayashi Y, et al. Effect of reduced-dose vs high-dose glucocorticoids added to rituximab on remission induction in ANCA-associated vasculitis: a randomized clinical trial. *JAMA*. 2021;325(21):2178–2187.
116. Jayne DRW, Merkel PA, Schall TJ, Bekker P, Group AS. Avacopan for the treatment of ANCA-associated vasculitis. *N Engl J Med*. 2021;384(7):599–609.
119. Hellmich B, Sanchez-Alamo B, Schirmer JH, et al. EULAR recommendations for the management of ANCA-associated vasculitis: 2022 update. *Ann Rheum Dis*. 2024;83(1):30–47.
120. Stone JH, Merkel PA, Spiera R, et al. Rituximab versus cyclophosphamide for ANCA-associated vasculitis. *N Engl J Med*. 2010;363(3):221–232.
122. Jones RB, Tervaert JW, Hauser T, et al. Rituximab versus cyclophosphamide in ANCA-associated renal vasculitis. *N Engl J Med*. 2010;363(3):211–220.
126. de Groot K, Harper L, Jayne DR, et al. Pulse versus daily oral cyclophosphamide for induction of remission in antineutrophil cytoplasmic antibody-associated vasculitis: a randomized trial. *Ann Intern Med*. 2009;150(10):670–680.
128. Jayne DR, Gaskin G, Rasmussen N, et al. Randomized trial of plasma exchange or high-dosage methylprednisolone as adjunctive therapy for severe renal vasculitis. *J Am Soc Nephrol*. 2007;18(7):2180–2188.
129. Walsh M, Collister D, Zeng L, et al. The effects of plasma exchange in patients with ANCA-associated vasculitis: an updated systematic review and meta-analysis. *BMJ*. 2022;376:e064604.
131. KDIGO 2024 clinical practice guideline for the management of antineutrophil cytoplasmic antibody (ANCA)-Associated vasculitis. *Kidney Int*. 2024;105(3S):S71–S116.
136. Guillevin L, Pagnoux C, Karras A, et al. Rituximab versus azathioprine for maintenance in ANCA-associated vasculitis. *N Engl J Med*. 2014;371(19):1771–1780.
144. Casal Moura M, Specks U, Tehranian S, et al. Maintenance of remission and risk of relapse in myeloperoxidase-positive ANCA-associated vasculitis with kidney involvement. *Clin J Am Soc Nephrol*. 2023;18(1):47–59.
146. Kauffmann M, Bobot M, Robert T, et al. Disease activity and adverse events in patients with ANCA-associated vasculitides Undergoing long-term dialysis. *Clin J Am Soc Nephrol*. 2021;16(11):1665–1675.
149. Wong K, Pitcher D, Braddon F, et al. Effects of rare kidney diseases on kidney failure: a longitudinal analysis of the UK National Registry of Rare Kidney Diseases (RaDaR) cohort. *Lancet*. 2024;403(10433):1279–1289.
152. Chapman GB, Farrah TE, Chapman FA, et al. Utility of interval kidney biopsy in ANCA-associated vasculitis. *Rheumatology*. 2022;61(5):1966–1974.
158. Odler B, Riedl R, Gauckler P, et al. Risk factors for serious infections in ANCA-associated vasculitis. *Ann Rheum Dis*. 2023;82(5):681–687.
161. Caravaca-Fontán F, Stevens K, Padrón M, et al. Sodium-glucose cotransporter 2 inhibition in primary and secondary glomerulonephritis. *Nephrol Dial Transplant*. 2024;39(2):328–340.
167. Hruskova Z, Stel VS, Jayne D, et al. Characteristics and outcomes of granulomatosis with polyangiitis (Wegener) and microscopic polyangiitis requiring renal replacement therapy: results from the European renal association-European dialysis and transplant association registry. *Am J Kidney Dis*. 2015;66(4):613–620.
174. Canney M, O'Hara PV, McEvoy CM, et al. Spatial and temporal clustering of anti-glomerular basement membrane disease. *Clin J Am Soc Nephrol*. 2016;11(8):1392–1399.
176. Cui Z, Zhao J, Jia XY, Zhu SN, Zhao MH. Clinical features and outcomes of anti-glomerular basement membrane disease in older patients. *Am J Kidney Dis*. 2011;57(4):575–582.
193. Cui Z, Zhao MH, Segelmark M, Hellmark T. Natural autoantibodies to myeloperoxidase, proteinase 3, and the glomerular basement membrane are present in normal individuals. *Kidney Int*. 2010;78(6):590–597.
195. Kuang H, Liu J, Jia XY, Cui Z, Zhao MH. Autoimmunity in anti-glomerular basement membrane disease: a review of mechanisms and prospects for immunotherapy. *Am J Kidney Dis*. 2023;81(1):90–99.
196. Pedchenko V, Bondar O, Fogo AB, et al. Molecular architecture of the Goodpasture autoantigen in anti-GBM nephritis. *N Engl J Med*. 2010;363(4):343–354.
198. Floyd L, Bate S, Hadi KA, et al. Risk stratification to predict renal survival in anti-glomerular basement membrane disease. *J Am Soc Nephrol*. 2023;34(3):505–514.
200. Shen CR, Jia XY, Luo W, et al. Laminin-521 is a novel target of autoantibodies associated with lung hemorrhage in anti-GBM disease. *J Am Soc Nephrol*. 2021;32(8):1887–1897.
204. van Daalen EE, Jennette JC, McAdoo SP, et al. Predicting outcome in patients with anti-GBM glomerulonephritis. *Clin J Am Soc Nephrol*. 2018;13(1):63–72.
212. McAdoo SP, Pusey CD. Anti-glomerular basement membrane disease. *Clin J Am Soc Nephrol*. 2017;12(7):1162–1172.
218. Uhlin F, Szpirt W, Kronbichler A, et al. Endopeptidase cleavage of anti-glomerular basement membrane antibodies. *J Am Soc Nephrol*. 2022;33(4):829–838.
222. Huart A, Josse AG, Chauveau D, et al. Outcomes of patients with Goodpasture syndrome: a nationwide cohort-based study from the French Society of Hemapheresis. *J Autoimmun*. 2016;73:24–29.
226. KDIGO 2021 clinical practice guideline for the management of glomerular diseases. *Kidney Int*. 2021;100(4S):S1–S276.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

REFERENCES

1. Kronbichler A, Bajema I, Geetha D, Säemann M. Novel aspects in the pathophysiology and diagnosis of glomerular diseases. *Ann Rheum Dis.* 2023;82(5):585–593.
2. Jennette JC. Rapidly progressive crescentic glomerulonephritis. *Kidney Int.* 2003;63(3):1164–1177.
3. van Daalen EE, Wester Trejo MAC, Göçeroğlu A, et al. Developments in the histopathological classification of ANCA-associated glomerulonephritis. *Clin J Am Soc Nephrol.* 2020;15(8):1103–1111.
4. Trivioli G, Gopaluni S, Urban ML, et al. Slowly progressive anti-neutrophil cytoplasmic antibody-associated renal vasculitis: clinico-pathological characterization and outcome. *Clin Kidney J.* 2021;14(1):332–340.
5. Rhee RL, Hogan SL, Poulton CJ, et al. Trends in long-term outcomes among patients with antineutrophil cytoplasmic antibody-associated vasculitis with renal disease. *Arthritis Rheumatol.* 2016;68(7):1711–1720.
6. Klaus R, Kanzelmeyer N, Haffner D, Lange-Sperandio B. Outcome of rituximab treatment in children with non-dialysis-dependent anti-GBM disease. *Pediatr Nephrol.* 2025;40(2):423–430.
7. Barbour SJ, Coppo R, Zhang H, et al. Application of the international IgA nephropathy prediction tool one or two years post-biopsy. *Kidney Int.* 2022;102(1):160–172.
8. Jennette JC, Falk RJ, Bacon PA, et al. 2012 revised international Chapel Hill consensus conference nomenclature of vasculitides. *Arthritis Rheum.* 2013;65(1):1–11.
9. Kronbichler A, Shin JI, Lee KH, et al. Clinical associations of renal involvement in ANCA-associated vasculitis. *Autoimmun Rev.* 2020;19(4):102495.
10. Kronbichler A, Bajema IM, Bruchfeld A, Mastroianni Kirsztajn G, Stone JH. Diagnosis and management of ANCA-associated vasculitis. *Lancet.* 2024;403(10427):683–698.
11. Lyons PA, Peters JE, Alberici F, et al. Genome-wide association study of eosinophilic granulomatosis with polyangiitis reveals genomic loci stratified by ANCA status. *Nat Commun.* 2019;10(1):5120.
12. Gisslander K, White A, Aslett L, et al. Data-driven subclassification of ANCA-associated vasculitis: model-based clustering of a federated international cohort. *Lancet Rheumatol.* 2024;6(11):e762–e770.
13. Frohnert PP, Sheps SG. Long-term follow-up study of periarteritis nodosa. *Am J Med.* 1967;43(1):8–14.
14. Nelveg-Kristensen KE, Szpirt W, Carlson N, et al. Increasing incidence and improved survival in ANCA-associated vasculitis-a Danish nationwide study. *Nephrol Dial Transplant.* 2021;37(1):63–71.
15. Kronbichler A, Leierer J, Gauckler P, Shin JI. Comorbidities in ANCA-associated vasculitis. *Rheumatology.* 2020;59(suppl 3):iii79–iii83.
16. Moiseev S, Bulanov N, Crnogorac M, et al. Traditional and disease specific risk factors for cardiovascular events in ANCA-associated vasculitis: a multinational retrospective study. *J Rheumatol.* 2023;50(8):1094.
17. Wallace ZS, Fu X, Harkness T, Stone JH, Zhang Y, Choi H. All-cause and cause-specific mortality in ANCA-associated vasculitis: overall and according to ANCA type. *Rheumatology.* 2020;59(9):2308–2315.
18. Ocon AJ, Reed G, Pappas DA, Curtis JR, Kremer JM. Short-term dose and duration-dependent glucocorticoid risk for cardiovascular events in glucocorticoid-naive patients with rheumatoid arthritis. *Ann Rheum Dis.* 2021;80(12):1522–1529.
19. Rahmattulla C, Mooyaart AL, van Hooven D, et al. Genetic variants in ANCA-associated vasculitis: a meta-analysis. *Ann Rheum Dis.* 2016;75(9):1687–1692.
20. Mohammad AJ, Mortensen KH, Babar J, et al. Pulmonary involvement in antineutrophil cytoplasmic antibodies (ANCA)-associated vasculitis: the influence of ANCA subtype. *J Rheumatol.* 2017;44(10):1458–1467.
21. Berti A, Warner R, Johnson K, et al. Brief report: circulating cytokine profiles and antineutrophil cytoplasmic antibody specificity in patients with antineutrophil cytoplasmic antibody-associated vasculitis. *Arthritis Rheumatol.* 2018;70(7):1114–1121.
22. Specks U, Merkel PA, Seo P, et al. Efficacy of remission-induction regimens for ANCA-associated vasculitis. *N Engl J Med.* 2013;369(5):417–427.
23. Walsh M, Merkel PA, Peh CA, et al. Plasma exchange and glucocorticoids in severe ANCA-associated vasculitis. *N Engl J Med.* 2020;382(7):622–631.
24. McGovern DP, McClure ME, Coates M, et al. Study protocol for a randomised, phase II, double-blind, experimental medicine study of obinutuzumab versus rituximab in ANCA-associated vasculitis: ObiVas. *BMJ Open.* 2024;14(7):e083277.
25. Suppiah R, Robson JC, Grayson PC, et al. 2022 American College of Rheumatology/European alliance of associations for Rheumatology classification criteria for microscopic polyangiitis. *Ann Rheum Dis.* 2022;81(3):321–326.
26. Robson JC, Grayson PC, Ponte C, et al. 2022 American College of Rheumatology/European alliance of associations for Rheumatology classification criteria for granulomatosis with polyangiitis. *Ann Rheum Dis.* 2022;81(3):315–320.
27. Deleted in review.
28. Puéchal X, Iudici M, Pagnoux C, et al. Comparative study of granulomatosis with polyangiitis subsets according to ANCA status: data from the French Vasculitis Study Group Registry. *RMD Open.* 2022;8(1).
29. Schirmer JH, Wright MN, Vonthein R, et al. Clinical presentation and long-term outcome of 144 patients with microscopic polyangiitis in a monocentric German cohort. *Rheumatology.* 2016;55(1):71–79.
30. Mohammad AJ. An update on the epidemiology of ANCA-associated vasculitis. *Rheumatology.* 2020;59(suppl 3):iii42–iii50.
31. Berti A, Cornec D, Crowson CS, Specks U, Matteson EL. The epidemiology of antineutrophil cytoplasmic autoantibody-associated vasculitis in olmsted county, Minnesota: a twenty-year US population-based study. *Arthritis Rheumatol.* 2017;69(12):2338–2350.
32. Watts RA, Hatemi G, Burns JC, Mohammad AJ. Global epidemiology of vasculitis. *Nat Rev Rheumatol.* 2022;18(1):22–34.
33. Pearce FA, Craven A, Merkel PA, Luqmani RA, Watts RA. Global ethnic and geographic differences in the clinical presentations of anti-neutrophil cytoplasm antibody-associated vasculitis. *Rheumatology.* 2017;56(11):1962–1969.
34. Weiner M, Bjørneklett R, Hrušková Z, et al. Proteinase-3 and myeloperoxidase serotype in relation to demographic factors and geographic distribution in anti-neutrophil cytoplasmic antibody-associated glomerulonephritis. *Nephrol Dial Transplant.* 2019;34(2):301–308.
35. Romagnani P, Kitching AR, Leung N, Anders HJ. The five types of glomerulonephritis classified by pathogenesis, activity and chronicity (GN-AC). *Nephrol Dial Transplant.* 2023;38(suppl 2):ii3–ii10.
36. Olson SW, Owshalimpur D, Yuan CM, et al. Relation between asymptomatic proteinase 3 antibodies and future granulomatosis with polyangiitis. *Clin J Am Soc Nephrol.* 2013;8(8):1312–1318.
37. Berglin E, Mohammad AJ, Dahlqvist J, et al. Anti-neutrophil cytoplasmic antibodies predate symptom onset of ANCA-associated vasculitis. A case-control study. *J Autoimmun.* 2021;117:102579.
38. Henderson SR, Horsley H, Frankel P, et al. Proteinase 3 promotes formation of multinucleated giant cells and granuloma-like structures in patients with granulomatosis with polyangiitis. *Ann Rheum Dis.* 2023;82(6):848–856.
39. Kitching AR, Anders HJ, Basu N, et al. ANCA-associated vasculitis. *Nat Rev Dis Prim.* 2020;6(1):71.
40. Söderberg D, Segelmark M. Neutrophil extracellular traps in ANCA-associated vasculitis. *Front Immunol.* 2016;7:256.
41. Xiao H, Schreiber A, Heeringa P, Falk RJ, Jennette JC. Alternative complement pathway in the pathogenesis of disease mediated by anti-neutrophil cytoplasmic autoantibodies. *Am J Pathol.* 2007;170(1):52–64.
42. Huugen D, van Esch A, Xiao H, et al. Inhibition of complement factor C5 protects against anti-myeloperoxidase antibody-mediated glomerulonephritis in mice. *Kidney Int.* 2007;71(7):646–654.
43. Xiao H, Dairaghi DJ, Powers JP, et al. C5a receptor (CD88) blockade protects against MPO-ANCA GN. *J Am Soc Nephrol.* 2014;25(2):225–231.
44. Hu P, Xiao H, Alba MA, et al. Myeloperoxidase-ANCA IgG induces different forms of small vessel vasculitis based on type of synergistic immune stimuli. *Kidney Int.* 2024;106(5):870–886.
45. Haas M, Eustace JA. Immune complex deposits in ANCA-associated crescentic glomerulonephritis: a study of 126 cases. *Kidney Int.* 2004;65(6):2145–2152.
46. O'Reilly VP, Wong L, Kennedy C, et al. Urinary soluble CD163 in active renal vasculitis. *J Am Soc Nephrol.* 2016;27(9):2906–2916.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

47. Aendekerk JP, Timmermans SAME, Busch MH, et al. Urinary soluble CD163 and disease activity in biopsy-proven ANCA-associated glomerulonephritis. *Clin J Am Soc Nephrol.* 2020;15(12):1740–1748.
48. Moran SM, Scott J, Clarkson MR, et al. The clinical application of urine soluble CD163 in ANCA-associated vasculitis. *J Am Soc Nephrol.* 2021;32(11):2920–2932.
49. Tam FW, Sanders JS, George A, et al. Urinary monocyte chemoattractant protein-1 (MCP-1) is a marker of active renal vasculitis. *Nephrol Dial Transplant.* 2004;19(11):2761–2768.
50. Moran SM, Monach PA, Zgaga L, et al. Urinary soluble CD163 and monocyte chemoattractant protein-1 in the identification of subtle renal flare in anti-neutrophil cytoplasmic antibody-associated vasculitis. *Nephrol Dial Transplant.* 2020;35(2):283–291.
51. Jayne DRW, Bruchfeld AN, Harper L, et al. Randomized trial of C5a receptor inhibitor avacopan in ANCA-associated vasculitis. *J Am Soc Nephrol.* 2017;28(9):2756–2767.
52. Brix SR, Stege G, Disteldorf E, et al. CC chemokine ligand 18 in ANCA-associated crescentic GN. *J Am Soc Nephrol.* 2015;26(9):2105–2117.
53. Sonnemann J, Klocke J, Bieringer M, et al. Urinary T cells identify renal antineutrophil cytoplasmic antibody-associated vasculitis and predict prognosis: a proof of concept study. *Kidney Int Rep.* 2023;8(4):871–883.
54. Prskalo L, Skopnik CM, Goerlich N, et al. Urinary CD4 + T cells predict renal relapse in ANCA-associated vasculitis. *J Am Soc Nephrol.* 2024;35(4):483–494.
55. Sreih AG, Cronin K, Shaw DG, et al. Diagnostic delays in vasculitis and factors associated with time to diagnosis. *Orphanet J Rare Dis.* 2021;16(1):184.
56. Boyle N, O'Callaghan M, Ataya A, et al. Pulmonary renal syndrome: a clinical review. *Breathe.* 2022;18(4):220208.
57. Hankard A, Puéchal X, Martin Silva N, et al. Characteristics of ANCA-associated vasculitis with aneurysms: case series and review of the literature. *Autoimmun Rev.* 2023;22(5):103293.
58. Dammacco R, Biswas J, Mohanan-Earatt A, et al. The eye is a common site of granulomatosis with polyangiitis. A collaborative study. *BMC Ophthalmol.* 2023;23(1):26.
59. Delaval L, Samson M, Schein F, et al. Temporal arteritis revealing antineutrophil cytoplasmic antibody-associated vasculitides: a case-control study. *Arthritis Rheumatol.* 2021;73(2):286–294.
60. Mukhtyar C, Lee R, Brown D, et al. Modification and validation of the Birmingham vasculitis activity score (version 3). *Ann Rheum Dis.* 2009;68(12):1827–1832.
61. McAdoo SP, Tanna A, Hrušková Z, et al. Patients double-seropositive for ANCA and anti-GBM antibodies have varied renal survival, frequency of relapse, and outcomes compared to single-seropositive patients. *Kidney Int.* 2017;92(3):693–702.
62. Crnogorac M, Horvatic I, Kacinari P, Ljubanovic DG, Galesic K. Serum C3 complement levels in ANCA associated vasculitis at diagnosis is a predictor of patient and renal outcome. *J Nephrol.* 2018;31(2):257–262.
63. Labrosse R, Barmettler S, Derfalvi B, et al. Rituximab-induced hypogammaglobulinemia and infection risk in pediatric patients. *J Allergy Clin Immunol.* 2021;148(2):523–532.e8.
64. Soussan M, Abisror N, Abad S, et al. FDG-PET/CT in patients with ANCA-associated vasculitis: case-series and literature review. *Autoimmun Rev.* 2014;13(2):125–131.
65. Dirikgil E, Tas SW, Verburgh CA, et al. Identifying relevant determinants of in-hospital time to diagnosis for ANCA-associated vasculitis patients. *Rheumatol Adv Pract.* 2022;6(2):rkac045.
66. Jennette JC, Nachman PH. ANCA glomerulonephritis and vasculitis. *Clin J Am Soc Nephrol.* 2017;12(10):1680–1691.
67. Chen M, Yu F, Wang SX, Zou WZ, Zhao MH, Wang HY. Antineutrophil cytoplasmic autoantibody-negative Pauci-immune crescentic glomerulonephritis. *J Am Soc Nephrol.* 2007;18(2):599–605.
68. Deleted in review.
69. Juanet C, Hassi I, Koirala A. ANCA-negative pauci-immune glomerulonephritis: a review. *Glomerular Dis.* 2024;4(1):189–199.
70. Ronsin C, Georges M, Chapelet-Debout A, et al. ANCA-negative pauci-immune necrotizing glomerulonephritis: a case series and a new clinical classification. *Am J Kidney Dis.* 2022;79(1):56–68.e1.
71. Jennette JC, Wilkman AS, Falk RJ. Anti-neutrophil cytoplasmic autoantibody-associated glomerulonephritis and vasculitis. *Am J Pathol.* 1989;135(5):921–930.
72. Harris AA, Falk RJ, Jennette JC. Crescentic glomerulonephritis with a paucity of glomerular immunoglobulin localization. *Am J Kidney Dis.* 1998;32(1):179–184.
73. Chalkia A, Thomas K, Giannou P, et al. Hypocomplementemia is associated with more severe renal disease and worse renal outcomes in patients with ANCA-associated vasculitis: a retrospective cohort study. *Ren Fail.* 2020;42(1):845–852.
74. Scaglioni V, Scolnik M, Catoggio LJ, et al. ANCA-associated pauci-immune glomerulonephritis: always pauci-immune? *Clin Exp Rheumatol.* 2017;35 suppl 103(1):55–58.
75. Hilhorst M, van Paassen P, van Rie H, et al. Complement in ANCA-associated glomerulonephritis. *Nephrol Dial Transplant.* 2017;32(8):1302–1313.
76. Neumann I, Regele H, Kain R, Birck R, Meisl FT. Glomerular immune deposits are associated with increased proteinuria in patients with ANCA-associated crescentic nephritis. *Nephrol Dial Transplant.* 2003;18(3):524–531.
77. Oba R, Kanzaki G, Sasaki T, et al. Long-term renal survival in anti-neutrophil cytoplasmic antibody-associated glomerulonephritis with complement C3 deposition. *Kidney Int Rep.* 2021;6(10):2661–2670.
78. Cassard A, Kounde C, Bouillet L, et al. Are serum C3 levels or kidney C3 deposits useful markers for predicting outcomes in patients with ANCA-associated vasculitis? *J Transl Autoimmun.* 2023;7:100217.
79. Hauer HA, Bajema IM, van Houwelingen HC, et al. Renal histology in ANCA-associated vasculitis: differences between diagnostic and serologic subgroups. *Kidney Int.* 2002;61(1):80–89.
80. Boudhabhay I, Delestre F, Coutance G, et al. Reappraisal of renal arteritis in ANCA-associated vasculitis: clinical characteristics, pathology, and outcome. *J Am Soc Nephrol.* 2021;32(9):2362–2374.
81. L'Imperio V, Vischini G, Pagni F, Ferraro PM. Bowman's capsule rupture on renal biopsy improves the outcome prediction of ANCA-associated glomerulonephritis classifications. *Ann Rheum Dis.* 2022;81(6):e95.
82. Berden AE, Jones RB, Erasmus DD, et al. Tubular lesions predict renal outcome in antineutrophil cytoplasmic antibody-associated glomerulonephritis after rituximab therapy. *J Am Soc Nephrol.* 2012;23(2):313–321.
83. Morita T, Suzuki Y, Churg J. Structure and development of the glomerular crescent. *Am J Pathol.* 1973;72(3):349–368.
84. Bronstein R, Pace J, Gowthaman Y, Salant DJ, Mallipattu SK. Podocyte-parietal epithelial cell interdependence in glomerular development and disease. *J Am Soc Nephrol.* 2023;34(5):737–750.
85. van Daalen EE, Neeskens P, Zandbergen M, et al. Podocytes and proteinuria in ANCA-associated glomerulonephritis: a case-control study. *Front Immunol.* 2019;10:1405.
86. Zimmermann M, Klaus M, Wong MN, et al. Deep learning-based molecular morphometrics for kidney biopsies. *JCI Insight.* 2021;6(7).
87. Haas M, Seshan SV, Barisoni L, et al. Consensus definitions for glomerular lesions by light and electron microscopy: recommendations from a working group of the Renal Pathology Society. *Kidney Int.* 2020;98(5):1120–1134.
88. Haas M, Mirocha J, Amann K, et al. Impact of consensus definitions on identification of glomerular lesions by light and electron microscopy. *Kidney Int Rep.* 2022;7(1):78–86.
89. Levy JB, Turner AN, Rees AJ, Pusey CD. Long-term outcome of anti-glomerular basement membrane antibody disease treated with plasma exchange and immunosuppression. *Ann Intern Med.* 2001;134(11):1033–1042.
90. Berden AE, Ferrario F, Hagen EC, et al. Histopathologic classification of ANCA-associated glomerulonephritis. *J Am Soc Nephrol.* 2010;21(10):1628–1636.
91. Brix SR, Noriega M, Tennstedt P, et al. Development and validation of a renal risk score in ANCA-associated glomerulonephritis. *Kidney Int.* 2018;94(6):1177–1188.
92. Bate S, McGovern D, Costigliolo F, et al. The improved kidney risk score in ANCA-associated vasculitis for clinical practice and trials. *J Am Soc Nephrol.* 2024;35(3):335–346.
93. Casal Moura M, Fervenza FC, Specks U, Sethi S. Kidney biopsy chronicity grading in antineutrophil cytoplasmic antibody-associated vasculitis. *Nephrol Dial Transplant.* 2022;37(9):1710–1721.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

94. Takeda R, Takahashi K, Kronbichler A, et al. Development of a kidney prognostic score in a Japanese cohort of patients with antineutrophil cytoplasmic autoantibody vasculitis. *Kidney Int Rep.* 2024;9(3):611–623.
95. Hauer HA, Bajema IM, de Heer E, Hermans J, Hagen EC, Bruijn JA. Distribution of renal lesions in idiopathic systemic vasculitis: a three-dimensional analysis of 87 glomeruli. *Am J Kidney Dis.* 2000;36(2):257–265.
96. Merindol J, Levraut M, Seitz-Polski B, Morand L, Martis N. Diagnostic significance of antineutrophil cytoplasmic antibody (ANCA) titres: a retrospective case-control study. *RMD Open.* 2023;9(2).
97. Antonelou M, Perea Ortega L, Harvey J, Salama AD. Anti-myeloperoxidase antibody positivity in patients without primary systemic vasculitis. *Clin Exp Rheumatol.* 2019;37(suppl 117(2)):86–89.
98. Ai S, Liu X, Chen G, et al. Characteristics and diagnostic challenge of antineutrophil cytoplasmic antibody positive infective endocarditis. *Am J Med.* 2022;135(11):1371–1377.
99. Herrmann K, Schinke S, Csernok E, Moosig F, Holle JU. Diagnostic value of procalcitonin in ANCA-associated vasculitis (AAV) to differentiate between disease activity, infection and drug hypersensitivity. *Open Rheumatol J.* 2015;9:71–76.
100. Konstantinov KN, Ulff-Møller CJ, Tzamaloukas AH. Infections and antineutrophil cytoplasmic antibodies: triggering mechanisms. *Autoimmun Rev.* 2015;14(3):201–203.
101. Deshayes S, Dolladille C, Dumont A, et al. A Worldwide Pharmacoepidemiologic update on drug-induced antineutrophil cytoplasmic antibody-associated vasculitis in the era of targeted therapies. *Arthritis Rheumatol.* 2022;74(1):134–139.
102. Han JY, Lee JM, Jung SY, et al. Comparison of agranulocytosis and anti-neutrophil cytoplasmic antibody-associated vasculitis caused by two antithyroid drugs: a pharmacovigilance study using the WHO international database. *Fundam Clin Pharmacol.* 2024;38(4):780–788.
103. Pendergraft WF, Niles JL. Trojan horses: drug culprits associated with antineutrophil cytoplasmic autoantibody (ANCA) vasculitis. *Curr Opin Rheumatol.* 2014;26(1):42–49.
104. Barnini C, Oni L, Kronbichler A. Course of paediatric ANCA-associated glomerulonephritis: advocating for an age-inclusive approach. *RMD Open.* 2024;10(3).
105. Vece TJ, Watkin LB, Nicholas S, et al. Copa syndrome: a novel autosomal dominant immune dysregulatory disease. *J Clin Immunol.* 2016;36(4):377–387.
106. Muñoz-Vahos CH, Herrera-Uribe S, Arbeláez-Cortés Á, et al. Clinical profile of levamisole-adulterated cocaine-induced vasculitis/vasculopathy: a 30-case series. *J Clin Rheumatol.* 2019;25(3):e16–e26.
107. Gill C, Sturman J, Ozbek L, et al. Cocaine-induced granulomatosis with polyangiitis-an under-recognized condition. *Rheumatol Adv Pract.* 2023;7(1):rkad027.
108. Stone JH, McDowell PJ, Jayne DRW, et al. The glucocorticoid toxicity index: measuring change in glucocorticoid toxicity over time. *Semin Arthritis Rheum.* 2022;55:152010.
109. Chanouzas D, McGregor JAG, Nightingale P, et al. Intravenous pulse methylprednisolone for induction of remission in severe ANCA associated Vasculitis: a multi-center retrospective cohort study. *BMC Nephrol.* 2019;20(1):58.
110. Furuta S, Nakagomi D, Kobayashi Y, et al. Effect of reduced-dose vs high-dose glucocorticoids added to rituximab on remission induction in ANCA-associated vasculitis: a randomized clinical trial. *JAMA.* 2021;325(21):2178–2187.
111. Nagle S, Nguyen Y, Guerry MJ, et al. Real-life use of the PEXIVAS reduced-dose glucocorticoid regimen in granulomatosis with polyangiitis and microscopic polyangiitis. *Ann Rheum Dis.* 2024.
112. Furuta S, Nakagomi D, Kobayashi Y, et al. Reduced-dose versus high-dose glucocorticoids added to rituximab on remission induction in ANCA-associated vasculitis: predefined 2-year follow-up study. *Ann Rheum Dis.* 2024;83(1):96–102.
113. Pepper RJ, McAdoo SP, Moran SM, et al. A novel glucocorticoid-free maintenance regimen for anti-neutrophil cytoplasm antibody-associated vasculitis. *Rheumatology.* 2019;58(2):260–268.
114. Casal Moura M, Gauckler P, Anders HJ, et al. Management of antineutrophil cytoplasmic antibody associated vasculitis with glomerulonephritis as proposed by the ACR 2021, EULAR 2022 and KDIGO 2021 Guidelines/Recommendations. *Nephrol Dial Transplant.* 2023;38(11):2637–2651.
115. Merkel PA, Niles J, Jimenez R, et al. Adjunctive treatment with avacopan, an oral C5a receptor inhibitor, in patients with antineutrophil cytoplasmic antibody-associated vasculitis. *ACR Open Rheumatol.* 2020;2(11):662–671.
116. Jayne DRW, Merkel PA, Schall TJ, Bekker P, Group AS. Avacopan for the treatment of ANCA-associated vasculitis. *N Engl J Med.* 2021;384(7):599–609.
117. Cortazar FB, Niles JL, Jayne DRW, et al. Renal recovery for patients with ANCA-associated vasculitis and low eGFR in the ADVOCATE trial of avacopan. *Kidney Int Rep.* 2023;8(4):860–870.
118. Windpessl M, Kostopoulou M, Conway R, et al. Preventing infections in immunocompromised patients with kidney diseases: vaccines and antimicrobial prophylaxis. *Nephrol Dial Transplant.* 2023;38(suppl 2):ii40–ii49.
119. Hellmich B, Sanchez-Alamo B, Schirmer JH, et al. EULAR recommendations for the management of ANCA-associated vasculitis: 2022 update. *Ann Rheum Dis.* 2024;83(1):30–47.
120. Stone JH, Merkel PA, Spiera R, et al. Rituximab versus cyclophosphamide for ANCA-associated vasculitis. *N Engl J Med.* 2010;363(3):221–232.
121. Unizony S, Villarreal M, Miloslavsky EM, et al. Clinical outcomes of treatment of anti-neutrophil cytoplasmic antibody (ANCA)-associated vasculitis based on ANCA type. *Ann Rheum Dis.* 2016;75(6):1166–1169.
122. Jones RB, Tervaert JW, Hauser T, et al. Rituximab versus cyclophosphamide in ANCA-associated renal vasculitis. *N Engl J Med.* 2010;363(3):211–220.
123. Cortazar FB, Muhsin SA, Pendergraft WF, et al. Combination therapy with rituximab and cyclophosphamide for remission induction in ANCA vasculitis. *Kidney Int Rep.* 2018;3(2):394–402.
124. Gulati K, Edwards H, Prendecki M, et al. Combination treatment with rituximab, low-dose cyclophosphamide and plasma exchange for severe antineutrophil cytoplasmic antibody-associated vasculitis. *Kidney Int Rep.* 2021;100(6):1316–1324.
125. Pepper RJ, McAdoo SP, Moran SM, et al. A novel glucocorticoid-free maintenance regimen for anti-neutrophil cytoplasm antibody-associated vasculitis. *Rheumatology.* 2019;58(2):373.
126. de Groot K, Harper L, Jayne DR, et al. Pulse versus daily oral cyclophosphamide for induction of remission in antineutrophil cytoplasmic antibody-associated vasculitis: a randomized trial. *Ann Intern Med.* 2009;150(10):670–680.
127. Aqeel F, Xu L, Tomar O, Duchen J, Li T, Geetha D. Practice patterns of induction therapy in severe anti-neutrophil cytoplasmic autoantibody-associated vasculitis. *Kidney Int Rep.* 2022;7(12):2704–2708.
128. Jayne DR, Gaskin G, Rasmussen N, et al. Randomized trial of plasma exchange or high-dosage methylprednisolone as adjunctive therapy for severe renal vasculitis. *J Am Soc Nephrol.* 2007;18(7):2180–2188.
128a. Fussner LA, Flores-Suárez LF, Cartin-Ceba R, et al. Alveolar hemorrhage in antineutrophil cytoplasmic antibody-associated vasculitis: Results of an international randomized controlled trial (PEXIVAS). *Am J Respir Crit Care Med.* 2024;209(9):1141–1151.
129. Walsh M, Collister D, Zeng L, et al. The effects of plasma exchange in patients with ANCA-associated vasculitis: an updated systematic review and meta-analysis. *BMJ.* 2022;376:e064604.
130. Zeng L, Walsh M, Guyatt GH, et al. Plasma exchange and glucocorticoid dosing for patients with ANCA-associated vasculitis: a clinical practice guideline. *BMJ.* 2022;376:e064597.
131. KDIGO 2024. Clinical practice guideline for the management of antineutrophil cytoplasmic antibody (ANCA)-Associated vasculitis. *Kidney Int.* 2024;105(3S):S71–S116.
132. Jayne D, Rasmussen N, Andrassy K, et al. A randomized trial of maintenance therapy for vasculitis associated with antineutrophil cytoplasmic autoantibodies. *N Engl J Med.* 2003;349(1):36–44.
133. Hiemstra TF, Walsh M, Mahr A, et al. Mycophenolate mofetil vs azathioprine for remission maintenance in antineutrophil cytoplasmic antibody-associated vasculitis: a randomized controlled trial. *JAMA.* 2010;304(21):2381–2388.
134. Alberici F, Smith RM, Jones RB, et al. Long-term follow-up of patients who received repeat-dose rituximab as maintenance therapy for ANCA-associated vasculitis. *Rheumatology.* 2015;54(7):1153–1160.
135. Kemna MJ, Damoiseaux J, Austen J, et al. ANCA as a predictor of relapse: useful in patients with renal involvement but not in patients with nonrenal disease. *J Am Soc Nephrol.* 2015;26(3):537–542.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

136. Guillevin L, Pagnoux C, Karras A, et al. Rituximab versus azathioprine for maintenance in ANCA-associated vasculitis. *N Engl J Med.* 2014;371(19):1771–1780.
137. Charles P, Terrier B, Perrodeau É, et al. Comparison of individually tailored versus fixed-schedule rituximab regimen to maintain ANCA-associated vasculitis remission: results of a multicentre, randomised controlled, phase III trial (MAINRITSAN2). *Ann Rheum Dis.* 2018;77(8):1143–1149.
138. Charles P, Perrodeau É, Samson M, et al. Long-term rituximab use to maintain remission of antineutrophil cytoplasmic antibody-associated vasculitis: a randomized trial. *Ann Intern Med.* 2020;173(3):179–187.
139. Smith RM, Jones RB, Specks U, et al. Rituximab versus azathioprine for maintenance of remission for patients with ANCA-associated vasculitis and relapsing disease: an international randomised controlled trial. *Ann Rheum Dis.* 2023;82(7):937–944.
140. Deleted in review.
141. Zonozi R, Cortazar FB, Jeyabalan A, et al. Maintenance of remission of ANCA vasculitis by rituximab based on B cell repopulation versus serological flare: a randomised trial. *Ann Rheum Dis.* 2024;83(3):351–359.
142. Karras A, Pagnoux C, Haubitz M, et al. Randomised controlled trial of prolonged treatment in the remission phase of ANCA-associated vasculitis. *Ann Rheum Dis.* 2017;76(10):1662–1668.
143. Podestà MA, Mescia F, Ricchiuto A, et al. Predictors of hypogammaglobulinemia in ANCA-associated vasculitis after a rituximab-based induction: a multicentre study. *Rheumatology.* 2023;62(8):2850–2854.
144. Casal Moura M, Specks U, Tehranian S, et al. Maintenance of remission and risk of relapse in myeloperoxidase-positive ANCA-associated vasculitis with kidney involvement. *Clin J Am Soc Nephrol.* 2023;18(1):47–59.
145. McDermott G, Fu X, Cook C, et al. The effect of achieving serological remission on subsequent risk of relapse, end-stage renal disease and mortality in ANCA-associated vasculitis: a target trial emulation study. *Ann Rheum Dis.* 2022;81(10):1438–1444.
146. Kauffmann M, Bobot M, Robert T, et al. Disease activity and adverse events in patients with ANCA-associated vasculitides undergoing long-term dialysis. *Clin J Am Soc Nephrol.* 2021;16(11):1665–1675.
147. de Joode AA, Sanders JS, Stegeman CA. Renal survival in proteinase 3 and myeloperoxidase ANCA-associated systemic vasculitis. *Clin J Am Soc Nephrol.* 2013;8(10):1709–1717.
148. Geetha D, Specks U, Stone JH, et al. Rituximab versus cyclophosphamide for ANCA-associated vasculitis with renal involvement. *J Am Soc Nephrol.* 2015;26(4):976–985.
149. Wong K, Pitcher D, Braddon F, et al. Effects of rare kidney diseases on kidney failure: a longitudinal analysis of the UK National Registry of Rare Kidney Diseases (RaDaR) cohort. *Lancet.* 2024;403(10433):1279–1289.
150. Wester TMAC, Floßmann O, Westman KW, et al. Renal relapse in antineutrophil cytoplasmic autoantibody-associated vasculitis: unpredictable, but predictive of renal outcome. *Rheumatology.* 2019;58(1):103–109.
151. Benichou N, Charles P, Terrier B, et al. Proteinuria and hematuria after remission induction are associated with outcome in ANCA-associated vasculitis. *Kidney Int.* 2023;103(6):1144–1155.
152. Chapman GB, Farrah TE, Chapman FA, et al. Utility of interval kidney biopsy in ANCA-associated vasculitis. *Rheumatology.* 2022;61(5):1966–1974.
153. Flossmann O, Berden A, de Groot K, et al. Long-term patient survival in ANCA-associated vasculitis. *Ann Rheum Dis.* 2011;70(3):488–494.
154. Sánchez Álamo B, Moi L, Bajema I, et al. Long-term outcomes and prognostic factors for survival of patients with ANCA-associated vasculitis. *Nephrol Dial Transplant.* 2023;38(7):1655–1665.
155. Robson JC, Díaz Encarnación MM, Verhoeven P, et al. Correspondence on 'EULAR recommendations for the management of ANCA-associated vasculitis: 2022 update' by Hellmich. *Ann Rheum Dis.* 2024;83(10):e20.
156. Maillet T, Goletto T, Beltramo G, et al. Usual interstitial pneumonia in ANCA-associated vasculitis: a poor prognostic factor. *J Autoimmun.* 2020;106:102338.
157. Kronbichler A, Kerschbaum J, Gopaluni S, et al. Trimethoprim-sulfamethoxazole prophylaxis prevents severe/life-threatening infections following rituximab in antineutrophil cytoplasm antibody-associated vasculitis. *Ann Rheum Dis.* 2018;77(10):1440–1447.
158. Odler B, Riedl R, Gauckler P, et al. Risk factors for serious infections in ANCA-associated vasculitis. *Ann Rheum Dis.* 2023;82(5):681–687.
159. Sanchez-Alamo B, Moi L, Bajema I, et al. Long-term outcome of kidney function in patients with ANCA-associated vasculitis. *Nephrol Dial Transplant.* 2024;39(9):1483–1493.
160. Säemann M, Kronbichler A. Call for action in ANCA-associated vasculitis and lupus nephritis: promises and challenges of SGLT-2 inhibitors. *Ann Rheum Dis.* 2022;81(5):614–617.
161. Caravaca-Fontán F, Stevens K, Padrón M, et al. Sodium-glucose cotransporter 2 inhibition in primary and secondary glomerulonephritis. *Nephrol Dial Transplant.* 2024;39(2):328–340.
162. Farrah TE, Melville V, Czopek A, et al. Arterial stiffness, endothelial dysfunction and impaired fibrinolysis are pathogenic mechanisms contributing to cardiovascular risk in ANCA-associated vasculitis. *Kidney Int.* 2022;102(5):1115–1126.
163. Filer AD, Gardner-Medwin JM, Thambyrajah J, et al. Diffuse endothelial dysfunction is common to ANCA associated systemic vasculitis and polyarteritis nodosa. *Ann Rheum Dis.* 2003;62(2):162–167.
164. Wallace ZS, Fu X, Liao K, et al. Disease activity, antineutrophil cytoplasmic antibody type, and lipid levels in antineutrophil cytoplasmic antibody-associated vasculitis. *Arthritis Rheumatol.* 2019;71(11):1879–1887.
165. Bramlage CP, Kröplin J, Wallbach M, et al. Management of cardiovascular risk factors in patients with ANCA-associated vasculitis. *J Eval Clin Pract.* 2017;23(4):747–754.
166. Morgan MD, Turnbull J, Selamet U, et al. Increased incidence of cardiovascular events in patients with antineutrophil cytoplasmic antibody-associated vasculitides: a matched-pair cohort study. *Arthritis Rheum.* 2009;60(11):3493–3500.
167. Hruskova Z, Stel VS, Jayne D, et al. Characteristics and outcomes of granulomatosis with polyangiitis (Wegener) and microscopic polyangiitis requiring renal replacement therapy: results from the European renal association-European dialysis and transplant association registry. *Am J Kidney Dis.* 2015;66(4):613–620.
168. Geetha D, Eirin A, True K, et al. Renal transplantation in antineutrophil cytoplasmic antibody-associated vasculitis: a multicenter experience. *Transplantation.* 2011;91(12):1370–1375.
169. Göçeroğlu A, Rahmattulla C, Berden AE, et al. The Dutch transplantation in vasculitis (DUTRAVAS) study: outcome of renal transplantation in antineutrophil cytoplasmic antibody-associated glomerulonephritis. *Transplantation.* 2016;100(4):916–924.
170. Hruskova Z, Tesar V, Geetha D. Renal transplantation in antineutrophil cytoplasmic antibody-associated vasculitis: current perspectives. *Kidney Blood Press Res.* 2020;45(2):157–165.
171. Cook C, Jorge A, Fu X, et al. Association of renal transplantation with reduced risk of myocardial infarction and ischemic stroke in ANCA-associated vasculitis: an observational cohort study. *Semin Arthritis Rheum.* 2021;51(6):1180–1185.
172. Bharati J, Yang Y, Sharma P, Jhaveri KD. Atypical anti-glomerular basement membrane disease. *Kidney Int Rep.* 2023;8(6):1151–1161.
173. Appel GB, Lau WL. Treatment of rapidly progressive glomerulonephritis in the elderly. *Blood Purif.* 2018;45(1–3):208–212.
174. Canney M, O'Hara PV, McEvoy CM, et al. Spatial and temporal clustering of anti-glomerular basement membrane disease. *Clin J Am Soc Nephrol.* 2016;11(8):1392–1399.
175. Dowsett T, Oni L. Anti-glomerular basement membrane disease in children: a brief overview. *Pediatr Nephrol.* 2022;37(8):1713–1719.
176. Cui Z, Zhao J, Jia XY, Zhu SN, Zhao MH. Clinical features and outcomes of anti-glomerular basement membrane disease in older patients. *Am J Kidney Dis.* 2011;57(4):575–582.
177. Fischer EG, Lager DJ. Anti-glomerular basement membrane glomerulonephritis: a morphologic study of 80 cases. *Am J Clin Pathol.* 2006;125(3):445–450.
178. Nelveg-Kristensen KE, Madsen B, McClure M, et al. Age- and time-dependent increases in incident anti-glomerular basement membrane disease: a nationwide cohort study. *Clin Kidney J.* 2024;17(1):sfad261.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

179. Robson KJ, Ooi JD, Holdsworth SR, Rossjohn J, Kitching AR. HLA and kidney disease: from associations to mechanisms. *Nat Rev Nephrol.* 2018;14(10):636–655.
179a. Durcan ME, Nair B, Anderton JG. Human leucocyte antigen-associated anti-glomerular basement membrane disease in siblings. *Clin Kidney J.* 2019;13(2):261–262.
180. Pokidysheva EN, Seeger H, Pedchenko V, et al. Collagen IV. *J Biol Chem.* 2021;296:100590.
181. Clatworthy MR, Wallin EF, Jayne DR. Anti-glomerular basement membrane disease after alemtuzumab. *N Engl J Med.* 2008;359(7):768–769.
182. El Yamani N, Cote G, Riopel J, et al. Pembrolizumab-induced anti-GBM glomerulonephritis: a case report. *Kidney Med.* 2023;5(8):100682.
183. Gu QH, Xie LJ, Jia XY, et al. Fever and prodromal infections in anti-glomerular basement membrane disease. *Nephrology.* 2018;23(5):476–482.
184. Prendecki M, Clarke C, Cairns T, et al. Anti-glomerular basement membrane disease during the COVID-19 pandemic. *Kidney Int.* 2020;98(3):780–781.
185. Sacker A, Kung V, Andeen N. Anti-GBM nephritis with mesangial IgA deposits after SARS-CoV-2 mRNA vaccination. *Kidney Int.* 2021;100(2):471–472.
186. Aguiar R, Abbas A, Horsfield C, Moutzouris DA. The Case | A 64-year-old woman with retroperitoneal fibrosis, acute kidney injury, and active urine sediment. *Kidney Int.* 2021;99(2):493–494.
187. Ivković V, Bajema IM, Kronbichler A. Beyond serology: is there still a value of kidney biopsy in anti-glomerular basement membrane disease? *Kidney Int Rep.* 2023;8(12):2495–2498.
188. Antonelou M, Henderson SR, Bhangal G, et al. Binding truths: atypical anti-glomerular basement membrane disease mediated by IgA anti-glomerular basement membrane antibodies targeting the α1 chain of type IV collagen. *Kidney Int Rep.* 2019;4(1):163–167.
189. Nasr SH, Collins AB, Alexander MP, et al. The clinicopathologic characteristics and outcome of atypical anti-glomerular basement membrane nephritis. *Kidney Int.* 2016;89(4):897–908.
190. Antonelou M, Horsley H, Heptinstall L, Harber M, Salama AD. Binding truths: seronegative anti-glomerular basement membrane disease mediated by IgM anti-glomerular basement membrane antibodies. *Kidney Int Rep.* 2023;8(5):1117–1120.
191. Lerner RA, Glassock RJ, Dixon FJ. The role of anti-glomerular basement membrane antibody in the pathogenesis of human glomerulonephritis. *J Exp Med.* 1967;126(6):989–1004.
192. Wieslander J, Barr JF, Butkowski RJ, et al. Goodpasture antigen of the glomerular basement membrane: localization to noncollagenous regions of type IV collagen. *Proc Natl Acad Sci U S A.* 1984;81(12):3838–3842.
193. Cui Z, Zhao MH, Segelmark M, Hellmark T. Natural autoantibodies to myeloperoxidase, proteinase 3, and the glomerular basement membrane are present in normal individuals. *Kidney Int.* 2010;78(6):590–597.
194. Olson SW, Arbogast CB, Baker TP, et al. Asymptomatic autoantibodies associate with future anti-glomerular basement membrane disease. *J Am Soc Nephrol.* 2011;22(10):1946–1952.
195. Kuang H, Liu J, Jia XY, Cui Z, Zhao MH. Autoimmunity in anti-glomerular basement membrane disease: a review of mechanisms and prospects for immunotherapy. *Am J Kidney Dis.* 2023;81(1):90–99.
196. Pedchenko V, Bondar O, Fogo AB, et al. Molecular architecture of the Goodpasture autoantigen in anti-GBM nephritis. *N Engl J Med.* 2010;363(4):343–354.
197. Philip R, Dumont A, Martin Silva N, de Boysson H, Aouba A, Deshayes S. ANCA and anti-glomerular basement membrane double-positive patients: a systematic review of the literature. *Autoimmun Rev.* 2021;20(9):102885.
198. Floyd L, Bate S, Hadi KA, et al. Risk stratification to predict renal survival in anti-glomerular basement membrane disease. *J Am Soc Nephrol.* 2023;34(3):505–514.
199. McCall AS, Bhave G, Pedchenko V, et al. Inhibitory anti-peroxidasin antibodies in pulmonary-renal syndromes. *J Am Soc Nephrol.* 2018;29(11):2619–2625.
200. Shen CR, Jia XY, Luo W, et al. Laminin-521 is a novel target of autoantibodies associated with lung hemorrhage in anti-GBM disease. *J Am Soc Nephrol.* 2021;32(8):1887–1897.
201. Kuang H, Shen CR, Jia XY, et al. Autoantibodies against laminin-521 are pathogenic in anti-glomerular basement membrane disease. *Kidney Int.* 2023;104(6):1124–1134.
202. Shin JI, Geetha D, Szpirt WM, Windpessl M, Kronbichler A. Anti-glomerular basement membrane disease (Goodpasture disease): from pathogenesis to plasma exchange to IdeS. *Ther Apher Dial.* 2022;26(1):24–31.
203. Hellmark T, Segelmark M. Diagnosis and classification of Goodpasture's disease (anti-GBM). *J Autoimmun.* 2014;48–49:108–112.
204. van Daalen EE, Jennette JC, McAdoo SP, et al. Predicting outcome in patients with anti-GBM glomerulonephritis. *Clin J Am Soc Nephrol.* 2018;13(1):63–72.
205. Alchi B, Griffiths M, Sivalingam M, Jayne D, Farrington K. Predictors of renal and patient outcomes in anti-GBM disease: clinicopathologic analysis of a two-centre cohort. *Nephrol Dial Transplant.* 2015;30(5):814–821.
206. Bombassei GJ, Kaplan AA. The association between hydrocarbon exposure and anti-glomerular basement membrane antibody-mediated disease (Goodpasture's syndrome). *Am J Ind Med.* 1992;21(2):141–153.
207. Troxell ML, Houghton DC. Atypical anti-glomerular basement membrane disease. *Clin Kidney J.* 2016;9(2):211–221.
208. Casian A, Jayne D. Management of alveolar hemorrhage in lung vasculitides. *Semin Respir Crit Care Med.* 2011;32(3):335–345.
209. Serisier DJ, Wong RC, Armstrong JG. Alveolar haemorrhage in anti-glomerular basement membrane disease without detectable antibodies by conventional assays. *Thorax.* 2006;61(7):636–639.
210. Flores JC, Taube D, Savage CO, et al. Clinical and immunological evolution of oligoanuric anti-GBM nephritis treated by haemodialysis. *Lancet.* 1986;1(8471):5–8.
211. Winkler A, Zitt E, Sprenger-Mähr H, Soleiman A, Cejna M, Lhotta K. SARS-CoV-2 infection and recurrence of anti-glomerular basement disease: a case report. *BMC Nephrol.* 2021;22(1):75.
212. McAdoo SP, Pusey CD. Anti-glomerular basement membrane disease. *Clin J Am Soc Nephrol.* 2017;12(7):1162–1172.
213. Ohlsson S, Herlitz H, Lundberg S, et al. Circulating anti-glomerular basement membrane antibodies with predominance of subclass IgG4 and false-negative immunoassay test results in anti-glomerular basement membrane disease. *Am J Kidney Dis.* 2014;63(2):289–293.
214. Shen CR, Jia XY, Cui Z, Yu XJ, Zhao MH. Clinical and immunological characteristics of patients with combined anti-glomerular basement membrane disease and IgA nephropathy. *Clin Kidney J.* 2023;16(9):1480–1488.
215. Bu L, Said SM, Herrera Hernandez L, et al. The characteristics of concurrent anti-glomerular basement membrane nephritis and membranous nephropathy. *Kidney Int Rep.* 2023;8(10):2164–2167.
216. Stanton MC, Tange JD. Goodpasture's syndrome (pulmonary haemorrhage associated with glomerulonephritis). *Australas Ann Med.* 1958;7(2):132–144.
216a. Lockwood CM, Rees AJ, Pearson TA, et al. Immunosuppression and plasma-exchange in the treatment of Goodpasture's syndrome. *Lancet.* 1976;1(7962):711–715.
217. Marques C, Carvelli J, Biard L, et al. Prognostic factors in anti-glomerular basement membrane disease: a multicenter study of 119 patients. *Front Immunol.* 2019;10:1665.
218. Uhlin F, Szpirt W, Kronbichler A, et al. Endopeptidase cleavage of anti-glomerular basement membrane antibodies. *J Am Soc Nephrol.* 2022;33(4):829–838.
219. Connelly-Smith L, Alquist CR, Aqui NA, et al. Guidelines on the use of therapeutic apheresis in clinical practice–evidence-based approach from the writing committee of the American society for apheresis: the ninth special issue. *J Clin Apher.* 2023;38(2):77–278.
220. Johnson JP, Moore J, Austin HA, Balow JE, Antonovych TT, Wilson CB. Therapy of anti-glomerular basement membrane antibody disease: analysis of prognostic significance of clinical, pathologic and treatment factors. *Medicine (Baltim).* 1985;64(4):219–227.
221. Cui Z, Zhao J, Jia XY, et al. Anti-glomerular basement membrane disease: outcomes of different therapeutic regimens in a large single-center Chinese cohort study. *Medicine (Baltim).* 2011;90(5):303–311.
222. Huart A, Josse AG, Chauveau D, et al. Outcomes of patients with Goodpasture syndrome: a nationwide cohort-based study from the French Society of Hemapheresis. *J Autoimmun.* 2016;73:24–29.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

223. Mörtzell HM, Weiner M, Sperker W, et al. Analyses of registry data of patients with anti-GBM and antineutrophil cytoplasmatic antibody-associated (ANCA) vasculitis treated with or without therapeutic apheresis. *Transfus Apher Sci*. 2021;60(6):103227.
224. Biesenbach P, Kain R, Derfler K, et al. Long-term outcome of anti-glomerular basement membrane antibody disease treated with immunoadsorption. *PLoS One*. 2014;9(7):e103568.
225. Touzot M, Poisson J, Faguer S, et al. Rituximab in anti-GBM disease: a retrospective study of 8 patients. *J Autoimmun*. 2015;60:74–79.
226. KDIGO 2021. Clinical practice guideline for the management of glomerular diseases. *Kidney Int*. 2021;100(4S):S1–S276.
227. Arzoo K, Sadeghi S, Liebman HA. Treatment of refractory antibody mediated autoimmune disorders with an anti-CD20 monoclonal antibody (rituximab). *Ann Rheum Dis*. 2002;61(10):922–924.
228. Jain R, Dgheim H, Bomback AS. Rituximab for anti-glomerular basement membrane disease. *Kidney Int Rep*. 2019;4(4):614–618.
229. Zhu M, Wang J, Le W, et al. Relationship between anti-GBM antibodies and kidney outcomes in patients with anti-GBM disease. *J Nephrol*. 2023;36(3):789–797.
230. Segelmark M, Hellmark T, Wieslander J. The prognostic significance in Goodpasture's disease of specificity, titre and affinity of anti-glomerular-basement-membrane antibodies. *Nephron Clin Pract*. 2003;94(3):c59–c68.
231. Choy BY, Chan TM, Lai KN. Recurrent glomerulonephritis after kidney transplantation. *Am J Transplant*. 2006;6(11):2535–2542.

Downloaded for Yuhsuan Tseng (yhtseng@vghks.gov.tw) at Kaohsiung Veterans General Hospital from ClinicalKey.com by Elsevier on July 05, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.